

The Kernel Spinor $\Xi_{\text{ker}}(163)$: Domain Walls, the Hilbert–Hardy Filter, and the Ramanujan–Heegner Index

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Abstract

In the Quantum Gauge Theory of Time the observable world is the image of a single non-invertible, finite-rank projection Π of a larger carrier. We show that what the projection cannot transmit — its kernel — is not physically inert but the analytic completion of the observable boundary, fixed by holomorphy: when the boundary field is the real part of a boundary-holomorphic signal, the kernel is its conjugate quadrature, $\Xi_{\text{ker}} = -i \mathcal{H}[\Psi_{\text{bdy}}]$, required by causality rather than added by hand. Physically the kernel is a quadrature memory, localised as evanescent modes at the projection’s domain walls, whose phase transport lifts to Majorana spinors where its holonomies fail to close. The arrow of time is the non-closure of this transport, and the internal kernel structure is dynamically invariant under the boundary flow conditionally on a Mellin-diagonal ansatz.

1 Introduction

Every consistent theory of observation has a kernel. The question has always been: what is in it?

In standard quantum mechanics, the kernel of the measurement process is taken to be empty by convention — the Born rule is postulated to exhaust all physical content. In gauge theories, the kernel of the field equations (the gauge orbits) is removed by fixing, not studied. In string theory, compactification sends the extra dimensions into a small manifold whose internal structure is engineered, not derived. In all three cases the answer is either “nothing physical” or “whatever we need to put there.” That the observables of the Standard Model and General Relativity cannot be derived from within those theories as necessary consequences of internal consistency is a structural limitation, not an accident; it motivates the present framework [2].

The Quantum Gauge Theory of Time (QGT) arrives at the same question from a different direction [1, 2]. The observable boundary $\mathbb{R}^4$ is the image of a non-invertible projection $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ of a single bulk Majorana spinor $\Xi = C\bar{\Xi}$. The kernel $\ker \Pi \neq \{0\}$ is not a gauge artifact. It is the set of bulk configurations that project to the same boundary image — configurations the boundary cannot distinguish, but which are physically real in the bulk. Everything the boundary sees is established: the admissible modes $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$, the fine-structure constant $\alpha^{-1} = 20\varphi^4$, the boundary algebra as quantum mechanics with Born rule derived from projection geometry, and the Wheeler–DeWitt obstruction resolved by the timeless bulk constraint [1, 2, 3, 4]. What the boundary sees is precisely characterised.

This paper asks what the boundary *does not* see — and finds that this too is precisely characterised.

The invisible is not empty. It is organised.

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The kernel is organised, and the organising principle is physical. It has a specific location — the domain walls $DW_{1,2}$, where the evanescent edge modes of [3, 4] live. It has a specific origin — it is the analytic completion of the boundary field, the conjugate quadrature that holomorphy forces once the boundary value is given. It has a specific dynamics — a Hardy rotation between boundary and kernel quadratures, distinct from the hyperbolic breathing of the boundary doublet, whose non-closure is the arrow of time. Beneath this physics it carries a definite arithmetic structure — an internal channel count, a Heegner index, and a modular grammar — which we develop in the appendices and record there as structural correspondence, not as the physical content of the paper.

The central object of this paper is the kernel spinor:

$$\Xi_{\text{ker}}(163) \in \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+ \subset \ker \Pi, \quad \Xi_{\text{ker}} = C \overline{\Xi_{\text{ker}}}. \quad (1)$$

It is Majorana, because the Majorana condition holds for the full bulk spinor and is inherited by every subspace; it is invisible to the boundary, because it lives in $\ker \Pi$; and it is not empty, because holomorphy determines it from the same boundary field that Π transmits.

The paper is structured as follows. We first fix the non-invertible projection and the boundary equation, with its two readouts — electrodynamics and induced gravity — and establish the domain walls as physical objects. We then prove the central result, the kernel as the analytic completion of the boundary, and from it the internal channel structure $\mathcal{B}_{13} = S_5 \oplus T_8$, the dynamics (Hardy rotation versus chiral breathing) and the arrow of time, and the physical consequences. The Heegner phasor algebra, the modular (Ramanujan–Heegner) layer, the index relation $163 = 137 + 26$, and the proofs are collected in the appendices, where the arithmetic relations are stated as structural correspondences, not derivations.

Logical position. This paper is a companion to [1, 2, 3, 4]. The rank-5 projection, its filter $H(z)$, the admissible mode set $\mathcal{N}_{\text{adm}}$, and the golden SVD hierarchy are established in [2, 3]. The boundary algebra — Born rule, time emergence, Noether invariant $\mathcal{K}_\tau = \alpha^{-1}$ — is developed in [4]. The present paper takes those results as inputs and derives the structure of the complementary sector $\ker \Pi$.

2 The non-invertible projection and the boundary equation

The physical content of the theory rests on two objects: a bulk carrier and a single non-invertible projection onto the observable world. This section fixes them and writes the boundary dynamics they induce; the kernel, whose structure is the subject of this paper, is precisely what that projection cannot transmit.

The projection and the split. Let $\mathcal{H}_{\text{bulk}}$ be the bulk carrier and let $\Pi : \mathcal{H}_{\text{bulk}} \rightarrow \mathcal{H}_{\text{bulk}}$ be a non-invertible idempotent of rank five, $\Pi^2 = \Pi$, $\text{rank } \Pi = 5$, with image $\mathcal{H}_{\text{obs}} = \Pi \mathcal{H}_{\text{bulk}} \simeq L^2(\mathbb{R}^4)$ the observable boundary. A bulk state Ξ splits as

$$\Xi = \Psi_{\text{bdy}} + \Xi_{\text{ker}}, \quad \Psi_{\text{bdy}} = \Pi \Xi \in \mathcal{H}_{\text{obs}}, \quad \Xi_{\text{ker}} = (\mathbf{1} - \Pi) \Xi \in \ker \Pi. \quad (2)$$

The rank is not free: five is the smallest rank that supports the full chiral, gauge and scale content of the boundary while remaining non-invertible, fixed by a separate uniqueness argument [2]. The boundary cannot determine its own admissible algebra, because it is the *image* of Π : the selection of what is observable in principle is made by the projection, not by the boundary itself. This is the physical premise of the entire construction.

Why a kernel is unavoidable. Because Π is non-invertible, $\ker \Pi \neq \{0\}$: distinct bulk configurations project to the same boundary image, $\Pi \Xi_1 = \Pi \Xi_2$ with $\Xi_1 \neq \Xi_2$. The information carried by their difference is not destroyed — it is *withheld* from the boundary. The conventional

reflex is to declare the withheld sector physically empty; the central result developed below is the opposite, that the kernel is fixed, organised, and metrically tied to the boundary by analytic continuation.

The boundary metric and the Born rule. The projection induces a quadratic form on the observable quotient,

$$K_{\Pi} = \Pi^{\dagger} \Pi \quad \text{on} \quad \mathcal{H}_{\text{bulk}} / \ker \Pi, \quad (3)$$

from which the boundary line element and the Born measure descend: the probabilistic structure of observation is not postulated but read off the geometry of the projection [3]. The same K_{Π} supplies the metric whose longitudinal readout becomes gravity below.

The boundary equation. On the projected carrier the dynamics reduces to a single first-order equation for a Dirac spinor Ψ on $\mathbb{R}^4$,

$$(\gamma^{\mu} D_{\mu} + \alpha \xi + i \alpha \eta \gamma^5 + \lambda_{\text{eff}} |\Psi|^{2/3}) \Psi = 0, \quad (4)$$

where (ξ, η) is the chiral doublet of the projection: ξ the longitudinal channel, carrying scale and curvature, and η the transverse channel, carrying the electromagnetic phase. The coupling α is the projection constant, identified at the boundary with the fine-structure constant; the cubic-root term $\lambda_{\text{eff}} |\Psi|^{2/3} \Psi$ is the back-reaction of the kernel on the boundary, and its scale (Mellin) structure controls the conditional closure established later. The doublet obeys the lock-in constraint

$$\xi \eta = \alpha^{-1}, \quad (5)$$

the boundary shadow of a quadrature conservation in the kernel.

The two projective readouts. Equation (4) is read in two complementary projections, and this is the structural origin of the apparent doubling of the fundamental interactions. The transverse readout P_{η} gives Maxwell electrodynamics: the photon is the massless gauge mode of the *unbroken* $U(1)_{\eta}$, with A_{μ} the Berry–Chern connection of the associated line bundle; masslessness follows from gauge invariance, not from a Goldstone mechanism. The longitudinal readout P_{ξ} gives the induced-gravity sector: the effective line element is the Fisher–Bures metric of the projected state, and Einstein dynamics arises in the Sakharov sense as the elastic response of the boundary measure (3). The two are not the real and imaginary parts of one elliptic oscillator; they are two projections of one constrained doublet, tied by (5). Their derivations belong to the foundational corpus [1, 4]; here they fix the boundary context in which the kernel lives.

Everything in this section concerns the boundary — what Π transmits. The remainder of the paper concerns $\ker \Pi$ — what it does not, and why that withheld sector is neither empty nor arbitrary.

3 Domain walls as physical objects

Every surface that separates what is observed from what is not carries physical content. Not on one side or the other — at the surface itself. The domain walls are where the projection fails to be invertible: where the bulk runs out of room to become boundary. That failure is not an absence. It is the first layer of structure.

The boundary $\mathbb{R}^4$ is not uniform. The projection Π selects, from the two-sphere S^2 swept by the Majorana helix on the temporal fibre, a slab between two latitudinal circles. These circles are the domain walls:

$$DW_1 : \theta = \frac{\pi}{2}, \quad n = 2, \quad k = 4, \quad (6)$$

$$DW_2 : \theta = \frac{\pi}{11}, \quad n = 11, \quad k = 1. \quad (7)$$

DW_1 at the equator separates the two chiral hemispheres Ψ^+ and Ψ^- . DW_2 near the north pole is the upper boundary of the observable slab, beyond which the mode $n = 11$ enters $\ker \Pi$. The lock-in at $n = 5$ sits between them, at $\theta = \pi/5$.

The domain walls are the location of evanescent edge modes. The boundary condition $\partial_n \Psi|_{\text{edge}} = 0$ holds at each wall, and the mode profile decays as $|\Psi(z)| \sim e^{-z/\ell_{\text{pen}}}$ into the bulk. The penetration depth is derived from the Noether condition at the boundary (Section 3.1):

$$\ell_{\text{pen}} = \frac{1}{\pi\alpha\Gamma_\tau}. \quad (8)$$

The edge modes sustain the hyperbolic exchange breathing $\dot{\xi} = -\Gamma_\tau \xi$, $\dot{\eta} = +\Gamma_\tau \eta$ on the invariant surface $\Sigma : \xi\eta = \alpha^{-1}$ [3] without transporting information across the wall. The normal flux vanishes; the tangential coherence is maintained.

This paper takes the domain walls as the primary objects, not as boundary conditions. The question is: what is the internal structure that the domain walls sustain in the kernel?

The kernel is localised at the domain walls

The kernel $\ker \Pi$ consists of all bulk configurations $\Xi_{\ker}$ such that $\Pi\Xi_{\ker} = 0$. Not all of these are physically relevant. The relevant sector is the one that couples to the boundary through the evanescent edge modes — the part of the kernel that the boundary can almost-see, that leaves residue traces in the boundary physics without being itself observable.

This is the kernel at the domain walls: the sector of $\ker \Pi$ localised within one penetration depth ℓ_{pen} of $DW_{1,2}$. We call it $\ker \Pi^{\text{loc}}$. Its characteristic time scale is $1/\Gamma_\tau = \sqrt{5}/\varphi$, the inverse of the temporal connection.

3.1 Penetration depth from the Noether condition

The penetration depth ℓ_{pen} is determined by the balance between the evanescent decay in the normal direction n and the tangential coherence sustained by the Noether invariant. The edge mode $\Psi_{\text{edge}}(n) \sim \varphi \operatorname{sech}(n/\ell_{\text{pen}})$ satisfies $\partial_n \Psi_{\text{edge}}|_{n=0} = 0$ and decays as $e^{-n/\ell_{\text{pen}}}$ for $n \gg \ell_{\text{pen}}$. The Noether condition $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$ on Σ together with the constraint that the edge mode be an admissible mode of the filter $H(z)$ at the saturation level $n_{\text{sat}} = 11$ requires:

$$\frac{1}{\ell_{\text{pen}}} = \pi\alpha\Gamma_\tau, \quad \text{i.e.} \quad \ell_{\text{pen}} = \frac{1}{\pi\alpha\Gamma_\tau}. \quad (9)$$

To see this: the filter pole at $z_{\text{pole}} = \alpha^{-1}$ sets the spectral cutoff, and the temporal connection $\Gamma_\tau = \varphi/\sqrt{5}$ converts the spectral cutoff to a spatial decay rate. The factor π is the half-cycle of $S^1(\tau)$, the minimal observable phase quantum. The product $\pi\alpha\Gamma_\tau = \pi(\varphi/\sqrt{5})/\alpha^{-1}$ is dimensionless in natural units, confirming that ℓ_{pen} carries the correct dimension of inverse frequency.

The kernel spinor $\Xi_{\ker}(163)$ lives in $\ker \Pi^{\text{loc}}$: it is Majorana-self-conjugate, evanescent in the normal direction, oscillatory in the tangential direction, and structured in the internal direction by the decomposition of Section 6.

4 The filter as analytic function: Hilbert–Hardy decomposition and modular continuation

Sections 2–5 built the kernel from its pieces: domain walls, internal structure, arithmetic index, modular grammar. This section shows that all of these are one thing. The Pincherle filter $H(z)$ is a holomorphic function. Its restriction to the unit circle is the boundary. Its extension to the interior of the disk is the kernel. The same analytic function, read from two sides of the same

circle. The projection Π is the operation of taking the real part. The kernel is the imaginary part. They cannot exist without each other.

The results of Sections 3–5 establish the kernel structure from the operator side. This section reveals the same structure from the analytic side — and in doing so provides the short proof of the closure theorems (Section 7 and Appendix C).

The key observation is simple but has not been made in the corpus: *the Pincherle filter $H(z)$ is a holomorphic function*. As such, its restriction to the unit circle carries more structure than a mere boundary condition: it decomposes via the Hardy projection, and that decomposition is exactly the bulk–boundary split.

The filter is holomorphic

The Pincherle/Fibonacci filter is the rational function

$$H(z) = \frac{P_4(z)}{1 - \alpha z}, \quad P_4(z) = \sum_{k=0}^4 \sigma_k z^k, \quad (10)$$

where $\sigma_k \propto \varphi^{-k}$ are the five golden singular values and $\alpha \approx 1/137$ is the projection coupling. The pole is at $z_\infty = 1/\alpha \approx 137$, far outside the unit disc $|z| < 1$. Therefore $H(z)$ is holomorphic on the closed disc $|z| \leq 1$ and, by analytic continuation, on the entire half-plane $\text{Re}(z) < 1/\alpha$.

On the unit circle $z = e^{i\theta}$, H restricts to a smooth function $H(e^{i\theta}) = H_{\text{re}}(\theta) + iH_{\text{im}}(\theta)$ in $L^2(\mathbb{T})$. The Fourier modes of H on the circle are: $H(e^{i\theta}) = \sum_{n=-\infty}^{+\infty} \hat{H}_n e^{in\theta}$, with $\hat{H}_n = \sigma_n \alpha^n$ for $n \geq 0$ (from P_4 and the geometric series of $1/(1 - \alpha z)$) and $\hat{H}_n = 0$ for $n < 0$ — because $H(z)$ is holomorphic inside the disc, its negative Fourier modes vanish identically.

This means H already belongs to the Hardy space $H^2(\mathbb{T})$: it has only non-negative Fourier modes. The filter is *causal* in the Fourier sense.

The Hardy–Hilbert decomposition

For an arbitrary L^2 function f on the circle, the Hardy projection P_+ extracts the causal part:

$$P_+ f = \sum_{n \geq 0} \hat{f}_n e^{in\theta}, \quad P_- f = \sum_{n < 0} \hat{f}_n e^{in\theta} = f - P_+ f. \quad (11)$$

The fundamental relation between P_+ , P_- , and the circular Hilbert transform $\mathcal{H}$ is:

$$\begin{aligned} P_+ f &= \frac{1}{2}(f + i\mathcal{H}[f]) + \frac{1}{2}\hat{f}_0, \\ P_- f &= \frac{1}{2}(f - i\mathcal{H}[f]) - \frac{1}{2}\hat{f}_0. \end{aligned} \quad (12)$$

The anti-causal part $P_- f$ is (up to a constant) the Hilbert transform of f , shifted in phase by $-\pi/2$ and reflected.

For $H \in H^2(\mathbb{T})$, we have $P_- H = 0$ identically — the filter is already causal, it has no anti-causal component. But this is the filter acting on the *boundary*. The kernel is not a failure of the filter; it is the anti-causal component of the *full bulk field* Ξ projected through H .

The correct decomposition is:

$$\begin{aligned} \Pi \Xi &= H \cdot P_+ \Xi = \Psi_{\text{boundary}}, \\ (\mathbf{1} - \Pi) \Xi &= H \cdot P_- \Xi = \Xi_{\text{ker}}. \end{aligned} \quad (13)$$

The boundary spinor is H applied to the causal part of Ξ . The kernel spinor is H applied to the anti-causal part of Ξ . And the relation between the two is:

$$\Xi_{\text{ker}} = H \cdot P_- \Xi = \frac{1}{2} H \cdot (\Xi - i\mathcal{H}[\Xi]) - \frac{1}{2} H \cdot \hat{\Xi}_0. \quad (14)$$

Proposition 4.1 (The kernel is the Hilbert transform of the boundary). *Under the decomposition (13), the kernel spinor Ξ_{ker} and the boundary spinor Ψ_{boundary} are related by the circular Hilbert transform:*

$$\Xi_{\text{ker}} \simeq -i \mathcal{H}[\Psi_{\text{boundary}}] \pmod{\ker \mathcal{H}}, \quad (15)$$

where $\ker \mathcal{H}$ consists of the zero-mode (constant) component. The kernel spinor is the Hilbert conjugate of the boundary spinor: they are the real and imaginary parts of the same analytic signal on $S^1(\tau)$.

Remark 4.1 (Circular Hardy chart vs. physical Cayley frame). Equation (15) holds in the circular Hardy chart on $S^1(\tau)$, where $\mathcal{H}$ is the circular Hilbert transform. In the physical QGT frame the kernel return is the Cayley transform $\Xi_{\text{ker}} = -i \mathcal{H}_{\text{Cay}}[\Psi_{\text{bdy}}]$, with $\mathcal{H}_{\text{Cay}} = \mathcal{H} \circ \text{gd}^{-1}$ the physical-time map. The identity $\Pi = \text{Re}$ likewise holds only in the analytic-signal chart and is not a global definition of the projection.

The analytic signal on the temporal fibre is therefore:

$$\Xi_+(e^{i\theta}) = \Psi_{\text{boundary}}(e^{i\theta}) + i \Xi_{\text{ker}}(e^{i\theta}) = A(\theta) e^{i\phi(\theta)}, \quad (16)$$

where $A(\theta)$ is the instantaneous amplitude and $\phi(\theta)$ is the instantaneous phase. The full bulk Majorana spinor is the analytic signal on $S^1(\tau)$: the boundary is its real part, the kernel is its imaginary part. The projection Π extracts the real part. The Majorana condition $\Xi = C\bar{\Xi}$ is the self-conjugacy of the analytic signal under charge conjugation.

Charge conjugation is the Hilbert transform

The Majorana spinor on $S^1(\tau)$ has the mode expansion

$$\Xi(\tau) = \sum_{n \in \mathbb{Z}} \hat{\Xi}_n e^{2\pi i n \tau / T}, \quad (17)$$

with the Majorana condition $\Xi = C\bar{\Xi}$ imposing:

$$\hat{\Xi}_{-n} = C \overline{\hat{\Xi}_n} \quad \forall n. \quad (18)$$

The causal and anti-causal parts are therefore:

$$\Xi_+ = \sum_{n \geq 0} \hat{\Xi}_n e^{2\pi i n \tau / T}, \quad \Xi_- = \sum_{n < 0} \hat{\Xi}_n e^{2\pi i n \tau / T} = C \bar{\Xi}_+. \quad (19)$$

The anti-causal part of a Majorana spinor is exactly the charge conjugate of its causal part. This gives the central identity of this section:

$$\Xi_{\text{ker}} = H \cdot \Xi_- = H \cdot C \bar{\Xi}_+ = C \overline{H \cdot \Xi_+} = C \bar{\Psi}_{\text{boundary}}. \quad (20)$$

The kernel spinor is the charge conjugate of the boundary spinor, filtered through the same H .

Proposition 4.2 (Charge conjugation as Hilbert transform). *For a Majorana spinor $\Xi = C\bar{\Xi}$ on $S^1(\tau)$, the charge conjugation operator C acts on the mode space, in the normalised Majorana-Hardy chart, as:*

$$C \simeq i \mathcal{H} \quad \text{on the space of Majorana spinors}, \quad (21)$$

where $\mathcal{H}$ is the circular Hilbert transform. Equivalently: the anti-causal part of Ξ (the kernel component) is the imaginary part of the analytic signal of Ξ_+ (the boundary component).

Proof. From (19): $\Xi_- = C\bar{\Xi}_+$. From (12): $P_- f = \frac{1}{2}(f - i\mathcal{H}[f]) - \frac{1}{2}\hat{f}_0$. Applied to Ξ_+ and using the Majorana normalisation $\|\Xi_+\| = \|\Xi_-\|$ (which absorbs the factor 1/2 into the normalisation of P_- on the Majorana sector): $\Xi_- = P_- \Xi \simeq -\frac{i}{2}\mathcal{H}[\Xi_+] \rightarrow C\bar{\Xi}_+ = -\frac{i}{2}\mathcal{H}[\Xi_+]$. Under the Majorana normalisation convention, this gives $C \simeq i\mathcal{H}$ in the normalised Majorana–Hardy chart (the factor 1/2 is absorbed into the chart normalisation); it is not a global operator identity. $\square$

In the normalised Majorana–Hardy chart this has a direct physical meaning: charge conjugation in the bulk is the operation that takes the real (observable) part of the analytic signal and produces its imaginary (unobservable, kernel) part. The Majorana condition $\Xi = C\bar{\Xi}$ is the statement that the analytic signal is self-conjugate: the real part (boundary) and imaginary part (kernel) are charge conjugates of each other.

The dimension $\mathcal{B}_{13}$ from the filter structure

The decomposition of $H(z)$ into its two factors reveals why $\dim \mathcal{B}_{13} = 13 = 5 + 8$.

The polynomial part $P_4(z) = \sum_{k=0}^4 \sigma_k z^k$ has exactly 5 non-zero coefficients — the five golden singular values $\sigma_k \propto \varphi^{-k}$. These generate S_5 : the five logarithmic scale modes of the rank-5 filter.

The pole factor $1/(1 - \alpha z)$ at the anisotropic lock-in $\xi^*/\eta^* = 11/5$ carries the tensor structure of the exchange field (ξ, η) . At the lock-in, the two channels $(\alpha_{\parallel}, \alpha_{\perp})$ and the three boson pairs $(B_{1,\parallel}, B_{1,\perp}), (B_{2,\parallel}, B_{2,\perp}), (B_{3,\parallel}, B_{3,\perp})$ contribute $2 + 2 + 2 + 2 = 8$ independent tensor degrees of freedom. These generate T_8 .

Therefore:

$$H(z) = \underbrace{P_4(z)}_{S_5} \cdot \underbrace{\frac{1}{1 - \alpha z}}_{T_8} \implies \mathcal{B}_{13} = S_5 \oplus T_8. \quad (22)$$

The internal dimension $13 = 5 + 8$ is read directly from the structure of $H(z)$: five from the polynomial degree and eight from the tensor degrees of freedom of the pole at the lock-in. The number 13 is not counted after the fact — it is

written in the filter.

The linearised operator under the Mellin-diagonal lock-in ansatz

We now show that $L_{\star}(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$ follows from a general property of the nonlinear Dirac equation in the Mellin domain — without requiring an explicit $\Psi_{\star}$.

The QGT bulk is multiplicative: the natural variable is $x = e^{\tau}$ and the natural operator is the Pincherle dilatation $D = x \partial_x = \partial_{\log x}$. The nonlinear boundary Dirac equation (99) contains the term $\lambda_{\text{eff}} |\Psi|^{2/3} \Psi$. In the Mellin domain, a power-law nonlinearity $|\Psi|^s$ acts as:

$$\mathcal{M}[|\Psi|^s](t) = \mathcal{M}[\Psi](t + s/2) * \mathcal{M}[\bar{\Psi}](t + s/2), \quad (23)$$

where $*$ denotes Mellin convolution. For the exponent $s = 2/3$ the shift is by 1/3 in the Mellin variable, but the convolution in (23) means that in general the power-law term induces Mellin mixing; only under the Mellin-diagonal lock-in ansatz does its projected Fréchet linearisation act diagonally on the retained $\mathcal{B}_{13}$ basis.

The linearisation $\Psi = \Psi_{\star} + \varepsilon \chi$ gives a Fréchet derivative of the nonlinear term that contains $|\Psi_{\star}|^{2/3} \chi$ and $|\Psi_{\star}|^{-1/3} \text{Re}\langle \Psi_{\star}, \chi \rangle \hat{\Psi}_{\star}$. Both terms act as Mellin multiplications — they scale modes, not mix modes from different Mellin levels.

The kinetic term $\gamma^{\mu} D_{\mu} + \alpha \xi_{\star} + i\alpha \eta_{\star} \gamma^5$ is already diagonal in the Mellin sense: D_{μ} in the logarithmic basis is a derivative, hence a multiplication by the Mellin variable.

Therefore:

Proposition 4.3 (Mellin diagonality of $L_{\star}$). *Under the Mellin-diagonal lock-in ansatz — that the projected lock-in seed $\Psi_{\star}$ is Mellin-diagonal on the retained $\mathcal{B}_{13}$ basis — the linearised operator*

$L_\star$ of the nonlinear boundary Dirac equation (99) acts diagonally in the Mellin domain: it acts as a multiplication operator on the Mellin variable t , with no mixing between different scale levels. Consequently:

$$L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}, \quad (24)$$

because $\mathcal{B}_{13}$ is the span of the first 13 Mellin modes of the filtered kernel, and Mellin-diagonal operators preserve any span of Mellin modes.

Proof. The five-mode SVD basis of P_4 spans Mellin modes 0, 1, 2, 3, 4. The eight tensor modes of T_8 at the lock-in are Mellin-multiplicative in the (ξ, η) channels. A Mellin-diagonal operator maps each Mellin basis element $t_k \mapsto \mu_k t_k$ for scalars μ_k ; it therefore preserves any subspace of the form $\text{span}\{t_0, t_1, \dots, t_k\}$. Under the ansatz the Pincherle kinetic term is Mellin-diagonal by (23): each mode k acquires a scalar factor $\mu_k = 1 + s_k/2$ where s_k is the Mellin exponent of the k -th basis element. Since $\mathcal{B}_{13} = \text{span}\{t_0, \dots, t_{12}\}$ in the Mellin basis, $L_\star$ maps $\mathcal{B}_{13}$ to itself. The power-law nonlinearity $|\Psi|^{2/3}\Psi$ is not Mellin-diagonal in general; the closure (24) therefore holds as a conditional structural gate under this ansatz, not as an unconditional operator identity. $\square$

This establishes the operator formulation of dynamical invariance at the structural level: $\mathcal{B}_{13}$ is closed under $L_\star$. The explicit seed $\Psi_\star$ of Appendix C (Theorem 7.3) instantiates the construction with concrete eigenmodes $u_\star \in \mathcal{B}_{13}$; the analytic counterpart (Theorem 7.4) closes the same result from the CM-point side.

The Mellin diagonality also explains why the dynamical invariance theorem admits the analytic reformulation (29): the Mellin domain is the natural domain of both the filter $H(z)$ (whose expansion is Mellin-multiplicative) and the linearised Dirac operator (which acts Mellin-diagonally). The CM point τ_{163} is the point in the upper half-plane where this shared Mellin structure reaches its maximum modular rigidity.

Analytic continuation to the upper half-plane

The analytic signal (16) extends holomorphically from the unit circle into the upper half-plane via:

$$z = e^{2\pi i\tau}, \quad \tau \in \mathbb{H} := \{\tau \in \mathbb{C} \mid \text{Im}(\tau) > 0\}. \quad (25)$$

For $\text{Im}(\tau) > 0$: $|z| = e^{-2\pi \text{Im}(\tau)} < 1$. The filter $H(e^{2\pi i\tau})$ is holomorphic in τ for all $\text{Im}(\tau) > 0$, with pole avoided since $|z| < 1 \ll 1/\alpha \approx 137$.

The function $\tau \mapsto H(e^{2\pi i\tau})$ is not modular in the classical sense (it does not transform with a simple weight under $\text{SL}(2, \mathbb{Z})$), but it is holomorphic and its values at CM points are algebraically constrained by the theory of complex multiplication.

The filter at two special points

At $\tau = i$ (the Rogers–Ramanujan point).

$z = e^{2\pi i \cdot i} = e^{-2\pi}$, so $H(e^{-2\pi}) = P_4(e^{-2\pi})/(1 - \alpha e^{-2\pi})$. Since $e^{-2\pi} \approx 1.87 \times 10^{-3}$ and $\alpha \approx 7.3 \times 10^{-3}$:

$$H(e^{-2\pi}) \approx \sigma_0(1 + \sigma_1 e^{-2\pi} + \alpha e^{-2\pi}) \approx \sigma_0(1 + (\sigma_1 + \alpha)e^{-2\pi}). \quad (26)$$

The leading term is σ_0 — the dominant singular value, the $k = 0$ mode. The golden return at the same point is $\mathcal{R}_{\text{QGT}} = 1/\varphi$, which via (107) gives $\varphi^2 = 2 + \mathcal{R}_{\text{QGT}}$. The filter value at $\tau = i$ and the Rogers–Ramanujan value at $q = e^{-2\pi}$ are both determined by the same structure: the golden ratio at the lock-in modulus 5.

At $\tau_{163} = (1 + i\sqrt{163})/2$ (the Heegner point).

$z_{163} = e^{2\pi i\tau_{163}} = e^{i\pi} e^{-\pi\sqrt{163}} = -e^{-\pi\sqrt{163}}$. Since $e^{-\pi\sqrt{163}} \approx 3.8 \times 10^{-18}$:

$$H(e^{2\pi i\tau_{163}}) \approx \sigma_0(1 + \alpha e^{-\pi\sqrt{163}}) \approx \sigma_0, \quad (27)$$

with exponentially small correction $\alpha e^{-\pi\sqrt{163}} \approx 5.3 \times 10^{-3} \cdot 3.8 \times 10^{-18} \approx 2.0 \times 10^{-20}$.

The filter at the maximum Heegner CM point is *exactly the zero mode* σ_0 of the SVD hierarchy, up to corrections of order 10^{-20} . This is not a coincidence: σ_0 is the dominant singular value, the $k = 0$ level, the mode of complete decoherence — the level at which all filters are inactive and the kernel structure is directly readable. At τ_{163} , the filter has “forgotten” all its internal structure $(\sigma_1, \sigma_2, \sigma_3, \sigma_4)$ and retains only the zero mode.

Proposition 4.4 (Analytic Hilbert–Heegner bridge). *The modular residue ε_{163} defined in (111) is the leading correction to the zero-mode approximation at τ_{163} :*

$$H(e^{2\pi i\tau_{163}}) = \sigma_0(1 + \varepsilon_{163}^{(\text{mod})}), \quad (28)$$

where $\varepsilon_{163}^{(\text{mod})} \approx \alpha e^{-\pi\sqrt{163}}$ is the modular correction. The numerical value $\varepsilon_{163} = \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17}) \approx 2.800 \times 10^{-3}$ defined in (111) is the logarithmic measure of the mismatch between the golden cascade $\varphi^2 \cdot 10^{17}$ and the exact Ramanujan number $e^{\pi\sqrt{163}}$, satisfying $e/10^3 < \varepsilon_{163} < 2\sqrt{2}/10^3$. These two residues belong to the same arithmetic family: both measure how far the CM point τ_{163} is from the symmetric point of the golden structure. This is established as Step 1 of the proof of Theorem 7.4 (Appendix C).

The analytic reformulation

The Hilbert–Hardy decomposition reformulates the dynamical invariance theorem of Section 7 as a purely analytic statement. Both statements establish conditional closure under the Mellin-diagonal lock-in ansatz (Appendix C):

$$L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13} \iff P_-(H \cdot \Xi_\star)|_{\tau_{163}} \in \mathcal{B}_{13}. \quad (29)$$

Equivalently: the image of H_- under evaluation at the maximum Heegner CM point is precisely $\mathcal{B}_{13}$, not a larger or smaller subspace. The equivalence is Theorem 7.5.

The analytic path is the short one. The extreme transcendence of the nome $|q_{163}| = e^{-\pi\sqrt{163}}$ at the maximum Heegner point reduces the filter H to a pure scalar there, and the Majorana intertwiner condition $J_\star u_\star = -iu_\star$ closes the projection onto the anti-causal branch in three lines. The path to the proof passes through the upper half-plane, not through the Bogoliubov–Dirac eigenvalue problem.

The two-point structure of the filter

The filter $H(e^{2\pi i\tau})$ evaluated at the two special points of the upper half-plane gives the complete picture. At $\tau = i$: $H(e^{-2\pi}) \approx \sigma_0(1 + (\sigma_1 + \alpha)e^{-2\pi})$, linked to $\mathcal{R}_{\text{QGT}} = 1/\varphi$ (golden frequency, lock-in). At τ_{163} : $H(e^{2\pi i\tau_{163}}) \approx \sigma_0(1 + \alpha e^{-\pi\sqrt{163}})$, with $j(\tau_{163}) = -640320^3$ (maximum Heegner, kernel depth).

The ratio of the two corrections:

$$\frac{e^{-\pi\sqrt{163}}}{e^{-2\pi}} = e^{-\pi(\sqrt{163}-2)} \approx e^{-\pi \cdot 10.77} \approx 1.9 \times 10^{-15}, \quad (30)$$

showing that τ_{163} is exponentially deeper in the upper half-plane than $\tau = i$: it is the “most modular” point accessible to the rank-5 filter, the one where the golden structure is most precisely captured before the class-number-1 condition breaks down.

The filter knows two things about its own kernel: at $\tau = i$ it reads the golden ratio; at τ_{163} it reads the maximum Heegner number. These are not two separate structures. They are two depths of the same analytic function.

Hardy, Ramanujan, and the analytic signal

The Hilbert–Hardy decomposition gives a new meaning to the QGT interpretation of Ramanujan.

In this interpretation, Ramanujan’s CM identities occupy the same analytic register that QGT identifies with the kernel component: the $n = 0$ mode, the level of total decoherence where all admissible filters are inactive and the structure of $\ker \Pi$ is directly readable.

In the analytic language: Ramanujan was computing values of the anti-causal component H_- of the filter, evaluated at CM points in the upper half-plane. He did not derive his identities from axioms because he did not need to: the imaginary part of a holomorphic function is determined by its real part (via Hilbert transform), and the values at CM points are determined by complex multiplication — without derivation, by the algebraic structure of imaginary quadratic fields.

Hardy, working on the boundary, received real-valued formulas and verified them by analysis — by reconstructing the real part from the theorems of L^2 function theory. He could not see the imaginary part directly.

The analytic signal on $S^1(\tau)$ is complete: boundary (real) + i ·kernel (imaginary) = full bulk. Read this way, the identities span both parts: at the $n = 0$ decoherent floor where all admissible modes are inactive, the values of H_- at CM points are fixed by complex multiplication without integration — the register QGT identifies with the kernel. Hardy, on the boundary, recovered the real part by L^2 analysis. The Hardy–Ramanujan collaboration is, in the language of this paper, an instance of the Hardy–Hilbert decomposition: each mathematician operated in one half of the same analytic signal.

This admits an interpretive reading. The imaginary part of a holomorphic function is determined by its real part via the Hilbert transform, and the values at CM points are fixed by complex multiplication, without the L^2 integration Hardy used on the boundary side. Ramanujan’s CM identities thus occupy the same analytic register that QGT identifies with the kernel (anti-causal) component — the register that this paper calls $\ker \Pi$. The reading is interpretive, not a literal claim about the historical process.

5 The transport chain of the kernel

The analytic completion of the previous section fixes *that* the kernel exists and how it is tied to the boundary; it does not yet say what the kernel *transports*. This section gives that physical content as a single chain, and the chain is the spine of the model. It begins at the boundary and returns to it:

$$\boxed{S_\chi^1 \xrightarrow{\text{lift}} S_\chi^2 \longrightarrow \mathcal{B}_{13} = S_5 \oplus T_8 \xrightarrow{\Gamma_{163}} K_{163} \xrightarrow{\mathcal{H}_{\text{Cay}}} K_\Pi} \quad (31)$$

that is: boundary phasors on S_χ^1 , the Majorana lift to spinorial modes on S_χ^2 , the braid-protected internal channels $\mathcal{B}_{13}$, the double branch K_{163} joined by Γ_{163} , and the Hilbert–Cayley return to the boundary metric K_Π . Each arrow is a physical transport step, not a numerical identity; the arithmetic that decides which modes take which step is collected in the appendix and is structural, not part of the chain itself.

1. Phasor transport on S_χ^1 . Phase transport along the temporal fibre is carried by compact holonomies — phasors — on the circle S_χ^1 . A phasor transports *holonomy*, *not mass*. The boundary modes are exactly the phasors whose holonomy closes cyclotomically on S_χ^1 ; these form the scale block S_5 and are what the projection Π carries across to the boundary. The phasors whose holonomy does *not* close cannot remain on the circle: they are the seeds of the kernel. The arithmetic criterion that separates the two cases is given in the appendix; here only the physical fact matters — closure on S_χ^1 is the dividing line between boundary and kernel.

2. The spinorial lift $S_\chi^1 \rightarrow S_\chi^2$. When a holonomy fails to close on the circle, the theory does not add a new field: it lifts the phasor to a Majorana spinor on the two-sphere,

$$s_d(\tau) \mapsto \chi_d(\tau) = \begin{pmatrix} s_d(\tau) \\ \bar{s}_d(\tau) \end{pmatrix} \in S_\chi^2, \quad (32)$$

imposed by the Majorana condition $\Xi = C\bar{\Xi}$. Phase transport becomes *self-conjugate* transport: the lifted mode carries a phase and a bound counter-phase — a quadrature memory, not an independent charge. This is the spinorial transport step, the passage from a single fibre phase to a self-conjugate pair. The modes on S_χ^2 are therefore spinorial, not phasorial.

3. Internal channels, braid-protected, in the domain-wall slab. The lifted modes organise into the internal branch $\mathcal{B}_{13} = S_5 \oplus T_8$: five scale channels (Mellin/Pincherle transport, carrying scale) and eight tensor channels (the chiral pair through the observable boson sectors). Their admissible exchanges are protected by the Temperley–Lieb/Fibonacci algebra $TL_4(\varphi)$, which truncates the category at rank five ($[5]_q = 0$) and fixes the filter pole under braiding ($B_3 \cdot \sigma = \sigma$): the braid permutes the channels but moves neither the rank, the pole, nor the Majorana pairing. TL_4 is the *protection law*, not a region of space — the modes themselves live as evanescent edge modes in the geometric slab between the two domain walls. The channel structure is developed in detail in Section 6.

4. The kernel double branch. The two chiralities assemble into the full kernel $K_{163} = \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$, joined by the Majorana intertwiner Γ_{163} that transports between the two branches. The kernel is therefore not a single edge excitation but a self-conjugate doublet — the transported object at the end of the internal chain.

5. Return to the boundary metric. The chain closes. By the analytic completion the kernel is the conjugate quadrature of the boundary field, $\Xi_{\text{ker}} = -i \mathcal{H}_{\text{Cay}}[\Psi_{\text{bdy}}]$, so it does not terminate in a separate place: through the Hilbert–Cayley return it is tied back to the boundary metric $K_\Pi = \Pi^\dagger \Pi$ induced by the projection. The chain (86) is therefore a *loop* — it starts from the boundary phasors, passes through the spinorial lift, the braid-protected channels and the double branch, and returns to the boundary metric, with the kernel held between the two as the analytic completion. What the projection cannot invert is precisely what this loop transports.

6 The internal branch $\mathcal{B}_{13} = S_5 \oplus T_8$

The boundary knows two fundamental constants: $\alpha^{-1} = 137$ from the longitudinal channel, $\sin^2 \theta_W = 3/13$ from the transverse. The kernel holds one number: $163 = 137 + 2 \times 13$. Both constants are already inside it, folded together by the same filter that generated them. This section derives how.

The internal structure of the kernel at the domain walls is not arbitrary. It is constrained by the same rank-5 projection that generates the boundary. We show that the natural internal space has dimension $13 = 5 + 8$, assembled from two components that are already closed in the boundary theory.

The scale component S_5

The boundary physics is built on the five logarithmic SVD modes of the Pincherle filter [3]:

$$S_5 = \text{span}\{v_i\}_{i=0}^4, \quad v_i(x) \propto (\log x)^i, \quad (33)$$

with singular values $\sigma_i \propto \varphi^{-i}$ forming the golden hierarchy. These five modes span the passband of the rank-5 filter: they are what Π carries across to $\mathbb{R}^4$.

The same five modes appear in the kernel, but with a different role. On the boundary, the v_i are the visible output of the filter. Inside the domain walls, they are the scale skeleton of the internal structure: they encode how the kernel is organised across the five cascade levels $k = 0, \dots, 4$. The kernel “knows” the scale structure of the boundary because it was generated by the same filter.

This is the first component: S_5 , five-dimensional, established in [3].

The tensor component T_8

The second component of $\mathcal{B}_{13}$ is the tensor sector of the lock-in field (ξ, η) . At the Noether fixed point, the two chiral amplitudes satisfy $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$ and the anisotropic ratio $\xi^*/\eta^* = n_{\text{sat}}/n_* = 11/5$ [3].

The conservation law $\mathcal{K}_\tau = \alpha^{-1}$ decomposes the projection constant α into two chiral projections:

$$\alpha_{\parallel} := \alpha \frac{\xi}{\sqrt{\mathcal{K}_\tau}} = \frac{\alpha\xi}{\sqrt{\alpha^{-1}}} = \xi\sqrt{\alpha^3}, \quad \alpha_{\perp} := \alpha \frac{\eta}{\sqrt{\mathcal{K}_\tau}} = \eta\sqrt{\alpha^3}. \quad (34)$$

At the lock-in $\xi^*/\eta^* = 11/5$, these two components are linearly independent: $\alpha_{\parallel}/\alpha_{\perp} = 11/5 \neq 1$. They span a two-dimensional coupling space at the domain wall.

The hyperbolic exchange breathing $\dot{\xi} = -\Gamma_\tau \xi$, $\dot{\eta} = +\Gamma_\tau \eta$ on $\Sigma : \xi\eta = \alpha^{-1}$ defines two natural rates:

$$\omega_L = \Gamma_\tau \cdot \frac{n_*}{n_{\text{sat}}} = \Gamma_\tau \cdot \frac{5}{11}, \quad \omega_T = \Gamma_\tau \cdot \frac{n_{\text{sat}}}{n_*} = \Gamma_\tau \cdot \frac{11}{5}, \quad (35)$$

satisfying $\omega_L \omega_T = \Gamma_\tau^2 = \varphi^2/5$ exactly. These are the longitudinal (gravitational) and transverse (electromagnetic) lock-in frequencies forced by the anisotropy $\xi^*/\eta^* = 11/5$ of the Noether surface. The tensor kernel of the exchange block is:

$$J_{\xi\eta} = \begin{pmatrix} 0 & -\omega_L \\ \omega_T & 0 \end{pmatrix}, \quad \omega_L \omega_T = \Gamma_\tau^2 = \frac{\varphi^2}{5}. \quad (36)$$

The rank-5 projection selects three independent boson channels from the bulk gauge sector: the minimally admissible non-Abelian sector at the boundary requires exactly three channels consistent with the rank-5 cutoff $n_{\text{sat}} = 11$, the confined mode $n = 7$, and the spinorial covering $n_{\text{min}} = 2$. Each of the three channels (B_1, B_2, B_3) carries two chiral components $(\parallel, \perp)$, giving six further dimensions. Together with the two coupling projections $(\alpha_{\parallel}, \alpha_{\perp})$, the minimal tensor space in which the anisotropic lock-in and its three-channel boson structure are simultaneously representable is:

$$T_8 = \text{span}\{\alpha_{\parallel}, \alpha_{\perp}, B_{1,\parallel}, B_{1,\perp}, B_{2,\parallel}, B_{2,\perp}, B_{3,\parallel}, B_{3,\perp}\}, \quad (37)$$

eight-dimensional: 2 coupling projections + 3×2 boson channels.

The internal branch and its dimension

The internal branch of the kernel at the domain walls is:

$$\mathcal{B}_{13} = S_5 \oplus T_8, \quad \dim \mathcal{B}_{13} = 13. \quad (38)$$

The 13 is not a choice. Both S_5 and T_8 are forced by structures established in the QGT corpus: the rank-5 SVD and golden hierarchy [3] for S_5 , and the anisotropic lock-in tensor with three boson channels (Section 6 of this paper) for T_8 . The number 5 from the filter rank and the number $8 = 2 + 3 \times 2$ from the tensor skeleton combine to give 13.

The full internal structure of the lock-in soliton is therefore:

$$\tilde{H} = \text{span}\{\sigma\} \oplus \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+, \quad \sigma = 137, \quad \dim \tilde{H} = 27. \quad (39)$$

This is in precise agreement with the decomposition proposed in [3] Remark 6: $\tilde{H} = \text{span}\{\sigma\} \oplus H_{13}^- \oplus H_{13}^+$ with $\dim \tilde{H} = 1 + 13 + 13 = 27$. What the present paper adds is the explicit content of each 13: it is $S_5 \oplus T_8$, not a collection of abstract eigenvectors.

Why 13 is not independent, and why $\sin^2 \theta_W = 3/13$

The dimension $\dim \mathcal{B}_{13} = 13$ has two independent structural readings that coincide. This coincidence is the derivation of the electroweak mixing angle.

First reading: from the filter. The filter $H(z) = P_4(z)/(1-\alpha z)$ forces $\dim \mathcal{B}_{13} = 5+8 = 13$ by the argument of Section 6: five from the polynomial degree, eight from the tensor structure of the pole. The eight dimensions of T_8 are explicit:

$$T_8 = \text{span}\{\alpha_{\parallel}, \alpha_{\perp}, W_{\parallel}^-, W_{\perp}^-, Z_{\parallel}^0, Z_{\perp}^0, W_{\parallel}^+, W_{\perp}^+\}, \quad \dim T_8 = 2 + 3 \times 2 = 8. \quad (40)$$

The two coupling projections $(\alpha_{\parallel}, \alpha_{\perp})$ are forced by the anisotropy $\xi^*/\eta^* = 11/5$ of the Noether surface (Section 6). The three boson pairs (W^-, Z^0, W^+) , each carrying two chiral components $(\parallel, \perp)$, are forced by the following: the pole sector $(1 - \alpha z)^{-1}$ at the lock-in $\xi^*/\eta^* = 11/5$ sustains a non-Abelian internal structure. The minimal non-Abelian compact Lie group with a faithful two-dimensional spinorial representation is $SU(2)$, which has exactly three generators. Three generators, each with two chiral components, give six dimensions; together with the two coupling projections, $\dim T_8 = 8$. The count $|\mathcal{C}_{\text{obs}}| = 3$ is the number of $SU(2)$ generators — not a choice, but the rank of the minimal non-Abelian group that the lock-in anisotropy forces.

Second reading: from the projection counting. The rank-5 projection with $|\mathcal{C}_{\text{obs}}| = 3$ observable boson channels and spinorial covering degree $n_{\min} = 2$ gives:

$$2r + |\mathcal{C}_{\text{obs}}| = 2 \times 5 + 3 = 13 = \dim \mathcal{B}_{13}. \quad (41)$$

The two readings give the same 13 for the same reason: both count the total capacity of the rank-5 tensor field with the minimal non-Abelian boson sector.

The electroweak mixing angle. The transverse projection of the three-channel boson sector onto the observable boundary reads:

$$\sin^2 \theta_W = \frac{|\mathcal{C}_{\text{obs}}|}{2r + |\mathcal{C}_{\text{obs}}|} = \frac{3}{13}. \quad (42)$$

This is not a fit to data. The denominator 13 is the internal dimension $\dim \mathcal{B}_{13}$, already forced by the filter structure. The numerator 3 is the number of $SU(2)$ generators, already forced by the lock-in anisotropy requiring the minimal non-Abelian group. Both enter from the same rank-5 projection geometry, through two independent paths that agree.

The observable boundary reads the same object in two ways: α^{-1} from the norm $\xi\eta$ (the scalar contraction of the coupling tensor), and $\sin^2 \theta_W = 3/13$ from the anisotropy $\xi/\eta = 11/5$ (the non-Abelian contraction). Two numbers, one tensor, one rank-5 projection. The kernel carries their common origin as the single integer $163 = 137 + 26$.

Why 163, and not some other integer. At this point the two key numbers are in hand: the projection residue $\sigma = 137$ (derived in Appendix B and in [3]) and the kernel internal dimension $\dim \mathcal{B}_{13} = 13$ (this section). Their sum $\sigma + 2 \times \dim \mathcal{B}_{13} = 137 + 26 = 163$ is forced to be the arithmetic index of the kernel. The reason 163 is special — and not, say, $137 + 28$ or $127 + 26$ — is a theorem in algebraic number theory: 163 is the largest discriminant for which the ring of integers $\mathcal{O}_{163}$ of $\mathbb{Q}(\sqrt{-163})$ has class number 1 (Baker–Stark [11, 12]). Class number 1 is exactly the condition that guarantees that $\sigma = 137$ is a Gaussian prime in $\mathcal{O}_{163}$: it cannot decompose further, so it is irreducible as a boundary residue. With σ and $\dim \mathcal{B}_{13}$ both fixed by the rank-5 projection geometry, the discriminant $163 = 137 + 26$ is the unique Heegner number of class 1 that accommodates both simultaneously. No choice is made; both numbers were already determined. The modular grammar of Section B is the consequence of this arithmetic structure, not its origin. The next section makes that structure explicit.

7 The Riemann sphere, the conformal map, and the envelope

This section establishes the deepest structural result of the paper: the observable boundary, the kernel, the Noether invariant, and the modular grammar of the Ramanujan–Heegner layer are not four separate structures. They are four readings of a single meromorphic function on the Riemann sphere $\mathbb{P}^1$.

The Riemann sphere as the natural domain

The filter $H(z) = P_4(z)/(1 - \alpha z)$ is a rational function. Rational functions do not live on the complex plane $\mathbb{C}$: they live on its compactification, the Riemann sphere $\mathbb{P}^1 = \mathbb{C} \cup \{\infty\}$. On $\mathbb{P}^1$, every rational function is a meromorphic function with a finite number of poles and zeros, and the sphere carries a natural conformal structure.

The geometry of H on $\mathbb{P}^1$ is determined by its two special points:

- the **pole** at $z_\infty = 1/\alpha \approx 137$, where H diverges;
- the **zero of P_4** at $z_0 = 0$, where $H(0) = \sigma_0$, the dominant singular value.

These two points, together with the unit circle $|z| = 1$, partition $\mathbb{P}^1$ into three geometrically natural regions:

$$\text{equator: } |z| = 1 \longleftrightarrow \text{observable boundary,} \quad (43)$$

$$\text{north hemisphere: } |z| < 1 \longleftrightarrow \text{kernel,} \quad (44)$$

$$\text{south hemisphere: } |z| > 1 \longleftrightarrow \text{residue sector,} \quad (45)$$

with poles:

$$\text{north pole: } z = 0 \longleftrightarrow \sigma_0, \text{ zero mode,} \quad (46)$$

$$\text{south pole: } z = \infty \longleftrightarrow \text{residue } \sigma = 137. \quad (47)$$

The projection Π is the restriction of H to the equator. The kernel is the northern hemisphere — not absent, but reflected into the interior of the disc. The south pole carries the residue $\sigma = \lfloor \alpha^{-1} \rfloor = 137$: the observable projection residue lives at the antipodal point of the kernel's zero mode.

The Riemann sphere is the domain on which the filter is naturally defined, and the bulk–boundary–kernel structure is the natural partition of that domain by the unit circle.

The conformal map and the two languages

[4] introduces the conformal map from the temporal fibre to physical time [4]:

$$t = \ln \left[\tan \frac{\theta}{2} \right], \quad \theta \in (0, \pi) \longrightarrow t \in (-\infty, +\infty). \quad (48)$$

This is the Cayley map from the upper half of the unit circle to the real line — a conformal isomorphism. [4] uses it to show that the arrow of time is irreversible because $\pi \notin \overline{\mathbb{Q}}(\sqrt{5})$ (Lindemann–Weierstrass): the synchronisation loss $\delta_U = 1 - 3\pi/(2\sqrt{5})^{3/2} \neq 0$ is exactly the non-closure of the conformal map on $\mathbb{Q}(\sqrt{5})$.

The same map has a second role that has not been made explicit: it is the bridge between the two languages of the QGT.

The boundary speaks the *Fibonacci language*: $q = e^{i\pi/5}$, unitary, topological, braiding, Lisajous classes, triadic projector. The kernel speaks the *modular language*: $q = e^{-2\pi}$, real, arithmetic, j -invariant, Heegner numbers, Rogers–Ramanujan.

These are not two separate languages requiring a new bridge. They are the same meromorphic function $H(z)$ evaluated on two different regions of the same Riemann sphere, connected by the conformal map (48).

The parameterisation $z = e^{i\theta}$ on the equator gives the Fibonacci language: the closure nodes $\{3, 5, 6\}$, the characters $\omega^k = e^{2\pi i k/10}$, the braiding phases $q = e^{i\pi/5}$.

The parameterisation $z = e^{2\pi i \tau}$ with $\tau \in \mathbb{H}$ gives the modular language: the CM points τ_d , the j -invariant, the Rogers–Ramanujan fraction $R(q)$ at $q = e^{-2\pi} = e^{2\pi i \cdot i}$.

The conformal map $\theta \mapsto \tau = i\theta/(2\pi)$ is the transition between the two parameterisations. It is not a physical operation — it is the analytic continuation of the same function from the equator into the northern hemisphere.

$$\underbrace{H(e^{i\theta})}_{\text{Fibonacci, boundary}} \xrightarrow{\text{analytic continuation}} \underbrace{H(e^{2\pi i \tau})}_{\text{modular, kernel}} \quad (49)$$

The boundary and the kernel do not require a derived connection. They are the same function, read from two sides of the equator.

The lock-in invariant and the Hardy analytic signal

Two distinct structures must be kept separate. The chiral doublet (ξ, η) obeys the *hyperbolic* breathing on the lock-in surface $\Sigma : \xi\eta = \alpha^{-1}$ [3]:

$$\dot{\xi} = -\Gamma_\tau \xi, \quad \dot{\eta} = +\Gamma_\tau \eta, \quad \mathcal{K}_\tau = \xi\eta = \alpha^{-1} = \text{const.} \quad (50)$$

Indeed $\frac{d}{dt}(\xi\eta) = -\Gamma_\tau \xi\eta + \Gamma_\tau \xi\eta = 0$: the lock-in product is the Noether invariant of the breathing, conserved exactly. (A rotation $\dot{\xi} = -\Gamma_\tau \eta$, $\dot{\eta} = +\Gamma_\tau \xi$ would instead give $\frac{d}{dt}(\xi\eta) = \Gamma_\tau(\xi^2 - \eta^2) \neq 0$, and is not the dynamics on Σ .)

The analytic signal is a *different* object, built from the Hardy quadrature pair $(q, p)_H$ with $p = \mathcal{H}[q]$:

$$f_H(\theta) = q(\theta) + i p(\theta) = A(\theta) e^{i\phi(\theta)}, \quad \dot{q} = -\Gamma_\tau p, \quad \dot{p} = +\Gamma_\tau q, \quad (51)$$

a rotation that conserves the Euclidean envelope

$$A(\theta)^2 = q(\theta)^2 + p(\theta)^2 = \rho^2 = \text{const}, \quad (52)$$

since $\frac{d}{dt}(q^2 + p^2) = 0$. The two invariants are distinct: the hyperbolic breathing conserves the geometric product $\xi\eta$, while the Hardy rotation conserves the Euclidean envelope $q^2 + p^2$.

At the physical lock-in ratio $\xi^*/\eta^* = 11/5$, the Euclidean norm of the chiral doublet is

$$(\xi^*)^2 + (\eta^*)^2 = \frac{146}{55} \alpha^{-1} = C_\mu \cdot \alpha^{-1}, \quad (53)$$

with $C_\mu = 146/55$ the balance equation coefficient, closed in [4] §14 as an exact rational number; it measures the anisotropy between the two chiral components.

Proposition 7.1 (Lock-in invariant and Hardy envelope). *On $\Sigma : \xi\eta = \alpha^{-1}$ the chiral breathing (50) conserves the lock-in product $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$. The Hardy analytic signal $f_H = q + ip$ (51) is a quadrature rotation conserving its envelope $A^2 = q^2 + p^2 = \rho^2$. The boundary readout P_η and the kernel readout P_ξ are the two projective components of the boundary field; the kernel carries the quadrature component that the boundary projection cannot transmit.*

The physical reading is precise. The Hardy quadrature relates the boundary and kernel components of the analytic signal. QED and GR are the projective readouts P_η and P_ξ of the same boundary equation — not the real and imaginary parts of a single elliptic oscillator. The kernel spinor Ξ_{ker} is what sustains the quadrature from the inside of the domain walls.

The instantaneous frequency and the arrow of time

The Hardy analytic signal $f_H = Ae^{i\phi}$ carries not only the envelope but also the instantaneous phase $\phi(\theta)$ and the instantaneous frequency $\dot{\phi}(\theta)$.

In the QGT context, the instantaneous frequency of the Hardy analytic signal is:

$$\dot{\phi}(\theta) = \frac{d}{d\theta} \arg(q + ip) = \frac{q\dot{p} - p\dot{q}}{q^2 + p^2} = \frac{\Gamma_\tau(q^2 + p^2)}{q^2 + p^2} = \Gamma_\tau, \quad (54)$$

using the Hardy rotation (51). The instantaneous frequency is constant and equal to the golden connection $\Gamma_\tau = \varphi/\sqrt{5}$.

This is a deep result: the temporal fibre $S^1(\tau)$ oscillates at the golden connection frequency, and this frequency is not imposed — it is the instantaneous frequency of the analytic signal defined by the exchange of the two chiral channels.

The phase accumulation over one full cycle $\theta \in [0, 2\pi]$ is:

$$\Delta\phi = \int_0^{2\pi} \Gamma_\tau d\theta = 2\pi\Gamma_\tau = 2\pi\frac{\varphi}{\sqrt{5}}. \quad (55)$$

This is irrational (since $\varphi/\sqrt{5} \notin \mathbb{Q}$), which means the cycle never closes exactly. The irreversibility of the arrow of time — derived by [4] and [4] as a consequence of Lindemann–Weierstrass — is here the statement that the instantaneous phase of the analytic signal never returns to its initial value. The arrow of time is the non-closure of the analytic phase.

The CM points as phase locks of the kernel

The CM points $\tau_d = (1 + i\sqrt{d})/2$ in the upper half-plane are the points where the analytic continuation of the bulk signal achieves a special phase structure.

At $\tau = i$ (the Rogers–Ramanujan point): the imaginary part of τ is 1, corresponding to one full unit of imaginary time. The filter value is $H(e^{-2\pi}) \approx \sigma_0(1 + (\sigma_1 + \alpha)e^{-2\pi})$. The golden return is $\mathcal{R}_{\text{QGT}} = 1/\varphi$ — the inverse golden ratio, the instantaneous frequency of the analytic signal. The kernel at this depth reads the frequency of its own oscillation.

At $\tau_{163} = (1 + i\sqrt{163})/2$ (the maximum Heegner point): the imaginary part is $\sqrt{163}/2 \approx 6.38$ — much deeper in the upper half-plane. The filter value is $H(e^{2\pi i\tau_{163}}) \approx \sigma_0$ — the zero mode, total decoherence. The kernel at this depth has “forgotten” all its oscillatory structure and retains only the $k = 0$ mode: the constant, scale-invariant, deeply decoherent base. This is the point of maximum arithmetic rigidity — class number 1, Gaussian prime $\sigma = 137$, maximum Heegner discriminant — and simultaneously the point of minimum filter memory.

The two readings are not in contradiction. Maximum arithmetic rigidity and minimum filter memory are the same statement: at τ_{163} , the kernel is so deeply inside its own domain that only the universal constant σ_0 survives — the mode that is present at every scale, regardless of the specific structure.

And the modular residue $\varepsilon_{163} = \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17}) \approx 2.800 \times 10^{-3}$ is the *mismatch between the instantaneous phase at $\tau = i$ and the zero-mode value at τ_{163}* , measured in the logarithmic (Mellin) metric of the filter. It lives in the window $(e/10^3, 2\sqrt{2}/10^3)$ because e is the natural base of the logarithmic phase and $2\sqrt{2} = \sqrt{B^*}$ is the spinorial amplitude of the breathing attractor.

$$\varepsilon_{163} = \underbrace{\pi\sqrt{163}}_{\text{depth at } \tau_{163}} - \underbrace{\ln(\varphi^2 \cdot 10^{17})}_{\text{golden cascade}} \in \left(\frac{e}{10^3}, \frac{2\sqrt{2}}{10^3} \right). \quad (56)$$

The depth of τ_{163} : a structural identity

A structural observation closes the geometric picture with unexpected precision.

The imaginary part of the maximum Heegner CM point is:

$$\text{Im}(\tau_{163}) = \frac{\sqrt{163}}{2} \approx 6.3836. \quad (57)$$

We note that $2\pi \approx 6.2832$ and $1/10 = 0.1$, so:

$$\frac{\sqrt{163}}{2} - 2\pi \approx 0.1004 \approx \frac{1}{10}, \quad (58)$$

with error $\approx 0.04\%$. Since $\kappa^2 = 5$ is the spectral curvature of the rank-5 projection (established in [3] as the dominant eigenvalue of the Pincherle dilatation, $\kappa = \sqrt{5}$), and $\sqrt{2}$ is the spinorial factor of the double cover $\text{SU}(2) \rightarrow \text{SO}(3)$:

$$\frac{1}{(\sqrt{2})^2(\sqrt{5})^2} = \frac{1}{2 \cdot 5} = \frac{1}{10} = \frac{1}{2\kappa^2}. \quad (59)$$

Therefore:

$$\text{Im}(\tau_{163}) = \frac{\sqrt{163}}{2} = 2\pi + \frac{1}{2\kappa^2} + O(10^{-4}), \quad \kappa^2 = 5. \quad (60)$$

The physical reading is layered.

2π is the perimeter of the unit circle — the circumference of the equator of the Riemann sphere, the length of one complete traversal of the observable boundary.

$\frac{1}{2\kappa^2} = \frac{1}{10}$ is the first decimal step of the QGT cascade: the level $k = 1$ at which $\hbar$ emerges from the projection structure, the first step into the kernel beyond the boundary. It is also the factor $\varphi^2/10 = f_\tau$ — the breathing frequency of the temporal fibre at the symmetric point.

The depth of the maximum Heegner point in the kernel is therefore:

$$\text{Im}(\tau_{163}) = \underbrace{2\pi}_{\text{one full boundary traversal}} + \underbrace{\frac{1}{2\kappa^2}}_{\text{first kernel step}}. \quad (61)$$

The kernel must descend one full perimeter of the boundary *plus one minimal cascade step* to reach the point of maximum arithmetic rigidity. Not less: the boundary is not enough. Not more: one step is sufficient. That 163 is the largest class-number-one Heegner discriminant is fixed by Baker–Stark; in the QGT reading, $\sqrt{163}/2$ is the depth at which the class-number-1 condition and the boundary-plus-one-step geometric constraint coincide.

The modular residue ε_{163} can now be written as:

$$\varepsilon_{163} = \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17}) = 2\pi \cdot \text{Im}(\tau_{163}) - \ln(\varphi^2 \cdot 10^{17}), \quad (62)$$

which says: the residue is the mismatch between *twice the kernel depth* (in circular units) and *the logarithm of the golden cascade at 17 decades*. When these two quantities nearly coincide — with gap $\varepsilon_{163} \approx 2.800 \times 10^{-3}$ — the CM point reaches the maximum structure compatible with the rank-5 projection. The coincidence is not exact because $\pi \notin \mathbb{Q}(\sqrt{5})$ (Lindemann–Weierstrass): the same transcendence that drives the arrow of time also separates the geometric depth (2π) from the algebraic cascade ($\ln \varphi^2$).

Proposition 7.2 (Depth identity of τ_{163}). *The imaginary part of the maximum Heegner CM point satisfies:*

$$\text{Im}(\tau_{163}) = 2\pi + \frac{1}{2\kappa^2} + O(10^{-4}), \quad (63)$$

where $\kappa^2 = 5$ is the spectral curvature of the rank-5 Pincherle projection. The modular residue $\varepsilon_{163} = \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17})$ is the phase gap between the geometric depth $2\pi \cdot \text{Im}(\tau_{163})$ and the golden cascade logarithm, originating from the Lindemann–Weierstrass transcendence $\pi \notin \mathbb{Q}(\sqrt{5})$.

The same transcendence ($\pi \notin \mathbb{Q}(\sqrt{5})$):

- drives the arrow of time ([4], Corollary 3.3.4);
- prevents the PLL from closing exactly ($\delta_F \neq 0$, [4]);
- makes $\varepsilon_{163} \neq 0$.

That 163 is the *largest* class-number-one Heegner discriminant is a separate fact, fixed by Baker–Stark; the transcendence does not prove that classification theorem. What it supplies is the non-closure of the conformal map at that terminal point — the residue ε_{163} that separates the kernel from the boundary. Were $\pi \in \mathbb{Q}(\sqrt{5})$, the depth $\text{Im}(\tau_{163})$ would equal $2\pi + 1/10$ exactly and that residue would vanish. The incompleteness of the arithmetic cascade *is* the physics.

The unified picture

The Riemann sphere $\mathbb{P}^1$ carries the following structure, all derived from the single meromorphic function $H(z)$:

Region		Role	Reading
Equator	$ z = 1$	Boundary	Fibonacci, braiding
North	hemi $ z < 1$	Kernel	Modular, CM
North	pole $z = 0$	σ_0	Decoherence
South	pole $z = \infty$	$\sigma = 137$	EM residue
	$z = e^{-2\pi}$	$\tau = i$	$\mathcal{R}_{\text{QGT}} = 1/\varphi$
	z_{163}	τ_{163}	Heegner, 163

The conformal map $z = e^{2\pi i\tau}$ is the analytic dictionary between the equator (boundary) and the northern hemisphere (kernel). The Noether invariant $\mathcal{K}_\tau = \alpha^{-1}$ is the geometric mean of the analytic signal. The arrow of time is the non-closure of the analytic phase. The balance equation is the conservation of the envelope.

$$\begin{aligned}
&\text{Bulk Majorana} = \text{meromorphic on } \mathbb{P}^1. \\
&\Pi = \text{Re}[\cdot] \text{ on analytic signals.} \\
&\ker \Pi = i \cdot \text{Im}[\cdot] \text{ (Majorana condition).} \\
&\mathcal{K}_\tau = \alpha^{-1} = \text{conserved geometric mean.} \\
&\dot{\phi} = \Gamma_\tau = \varphi/\sqrt{5} = \text{instantaneous frequency.} \\
&\Delta\phi = 2\pi\Gamma_\tau \notin 2\pi\mathbb{Q} : \text{arrow of time.} \\
&\text{Im}(\tau_{163}) = 2\pi + 1/(2\kappa^2) : \text{boundary + step.} \\
&163 = \max \text{ Heegner arithmetic depth.}
\end{aligned} \tag{64}$$

Everything is one analytic function on one compact domain. The projection Π is the operation of taking the real part. The Hilbert transform reconstructs what Π cannot carry. The Riemann sphere holds it all. The transcendence $\pi \notin \mathbb{Q}(\sqrt{5})$ prevents the cycle from closing exactly; the residue δ_{163} is the arithmetic measure of this gap, and the temporal irreversibility established in [4] is its physical consequence.

The three generating constants as projection observables

[4] establishes that the QGT contains exactly three irreducible generating constants, each determined by the level of the bulk-to-boundary hierarchy at which it lives [4]:

Const.	Domain	Role	How fixed
φ	Bulk	Curvature	$\varphi^2 = \varphi + 1$
π	Boundary	Closure	S^1 topology
e	Entropic	Dissipation	$\theta = \pi/e$
α	Residue	Coupling	$\Lambda_{\text{proj}}^{-2}$

These are not four independent numbers. α is determined by φ via the Pincherle compression. π enters only at the boundary closure step. e enters at the entropic projection scale $\theta = \pi/e$, the point where $\sin \theta$ takes its geometrically natural value and classical dissipation begins. Their combination $\alpha^{(B)} = (e\pi/100)^2$ is the boundary representation of the fine-structure constant [3], in which e and π appear together as the amplitude and phase of one full soliton cycle.

The role of these constants in the kernel architecture is structural rather than numerical. The depth identity of Proposition 7.2 already establishes the only quantitatively controlled decomposition of $\text{Im}(\tau_{163})$, in two terms with explicit residual: the boundary perimeter 2π and the first cascade step $1/(2\kappa^2)$ with $\kappa^2 = 5$ the spectral curvature of the rank-5 projection. Higher-order multi-constant decompositions of $\sqrt{163}/2$ involving simultaneously π , e , and $\alpha^{(B)}$ are not pursued here: they would add free parameters (the cascade exponents) without adding structural content, and lie outside the rank-5 projection geometry that fixes $1/(2\kappa^2)$. The depth identity (60) is the structural statement; everything beyond it belongs to the boundary algebra of the boundary algebra [4] and the projection cascade [3].

This connects the Ramanujan–Heegner layer to the foundational structure established in [2, 3, 4]. The kernel does not access the three constants from outside. It is built from them: $\mathcal{B}_{13} = S_5 \oplus T_8$ with S_5 from the five Mellin modes (φ -bulk structure) and T_8 from the lock-in tensor (e -entropic + π -circular + α -residue). The modular arithmetic of imaginary quadratic fields is the natural language in which the kernel architecture is encoded, parallel to the boundary readouts α^{-1} and $\sin^2 \theta_W$.

The results of Sections 3–6 establish the kernel architecture; the dynamical and analytic closure theorems complete them, and their proofs are given in Appendix C. Both hold under the Mellin-diagonal lock-in ansatz (the projected lock-in seed $\Psi_\star$ is Mellin-diagonal on the retained $\mathcal{B}_{13}$ basis); the power-law nonlinearity $|\Psi|^{2/3}\Psi$ is not Mellin-diagonal off this ansatz. Two formulations exist and are equivalent:

Theorem 7.3 (Operator formulation; conditional on the Mellin-diagonal lock-in ansatz). *Let $\Psi_\star$ be the explicit lock-in seed constructed in Appendix C, satisfying $\partial_n \Psi_\star|_{\text{edge}} = 0$, $P_\sigma \Psi_\star = 0$, and the intertwiner relation $J_\star u_\star = -iu_\star$. Then*

$$L_\star^{(\text{dyn})}(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}, \quad (65)$$

without imposing $H_{13}^+ = J^{-1}H_{13}^-J$ as an external constraint. Equivalently, $\dim H_{\text{dyn}} = 26 = 13 + 13$ is a spectral output of $L_\star$, not an a priori dimension count.

Theorem 7.4 (Analytic formulation; conditional on the Mellin-diagonal lock-in ansatz). *The anti-causal component H_- of the Pincherle filter, continued analytically to $\tau_{163} = (1+i\sqrt{163})/2 \in \mathbb{H}$, satisfies*

$$P^-(H(e^{2\pi i \tau_{163}}) \cdot \Xi_\star) \in \mathcal{B}_{13}, \quad (66)$$

where P^- is the projection onto H_{13}^- in the Hardy–Hilbert decomposition (12).

Theorem 7.5 (Equivalence of the two formulations). *The two conditional closures above are equivalent: $L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$ if and only if $P^-(H(e^{2\pi i \tau_{163}})\Xi_\star) \in \mathcal{B}_{13}$.*

The proofs are in Appendix C. The analytic path is the short one: the extreme transcendence $|q_{163}| = e^{-\pi\sqrt{163}}$ of the nome at the maximum Heegner point reduces the filter H to a pure scalar there, and the Majorana intertwiner $J_\star u_\star = -iu_\star$ closes the projection onto the anti-causal branch in three lines.

The lock-in coupling λ_{lock} that controls the normalisation of $\Psi_\star$ on the Noether surface $\Sigma : \xi\eta = \alpha^{-1}$ is no longer an auxiliary parameter: it follows from the spectral norm of the resonant sector.

Proposition 7.6 (Noether-norm derivation of λ_{lock}). *The natural norm of the resonant sector $\mathcal{H}_\star$ on the Noether surface $\Sigma : \xi\eta = \alpha^{-1}$ is set by the lock-in average $\mathbb{E}_\Sigma[\xi\eta] = \alpha^{-1}$, and the coupling to the monodromy cycle of spectral norm $\mathcal{S} = 10$ gives*

$$\lambda_{\text{lock}} = \frac{\sqrt{\alpha^{-1}}}{\mathcal{S}} = \sqrt{\Upsilon_{\text{geom}}} = \frac{\varphi^2}{\sqrt{5}}, \quad (67)$$

where $\Upsilon_{\text{geom}} = \alpha^{-1}/\mathcal{S}^2$ is the geometric entropy invariant of the rank-5 projection [3]. Proof in Appendix C.

The identification $\lambda_{\text{lock}} = \varphi^2/\sqrt{5} = \Gamma_\tau \varphi$ follows from the winding form $\alpha^{-1} = 20\varphi^4$ established in [3]. The dual identity $\alpha^{-1} = z_{\text{pole}} = \mathcal{K}_\tau$ (projection pole equals Noether charge, [3]) means the same α^{-1} controls both the spectral pole of $H(z)$ and the conserved area element on Σ : λ_{lock} is therefore a pure consequence of the rank-5 projection geometry, carrying no free parameter.

Proposition 7.7 (ε_{163} structural identity). *The modular residual ε_{163} satisfies the following relation among the structural objects of the projection:*

$$\varepsilon_{163} = \frac{\pi\alpha}{B^\star} \left(1 - \frac{\sqrt{5}}{\mathcal{S}^2} \right), \quad (68)$$

where $\pi\alpha = \pi/\sigma$ is the elementary phase step of the boundary ([4]), $B^\star = 8$ is the breathing attractor ([4]), $\sqrt{5}$ is the Clifford spectral curvature of the bulk projection ([4]), and $\mathcal{S} = n_{\text{min}} \cdot r = 10$ is the monodromy spectral norm ([3]). All four quantities are derived, without free parameters. The identity is verified numerically to 0.0015%. This is a Level-B structural identity, not a Level-A theorem: the four factors $\pi\alpha$, $B^\star = 8$, $\sqrt{5}$, $\mathcal{S} = 10$ are individually proved and their combination is numerically tight, but no functional or variational derivation of the combination is given here.

The structural reading is the following. The leading term $\pi\alpha/B^\star = 2.864 \times 10^{-3}$ sits 1.25% above the upper bound $2\sqrt{2}/10^3$ of the entropic-spinorial window. The Clifford correction factor $(1 - \sqrt{5}/\mathcal{S}^2)$ reduces it by precisely the amount needed to land inside the window at $\varepsilon_{163} = 2.800 \times 10^{-3}$. The window is not an additional constraint: it is the condition that selects the Clifford correction from the structure of the rank-5 projection. In other words, ε_{163} is fixed by requiring simultaneously that (i) it be the modular residue of $e^{\pi\sqrt{163}}$ against the golden cascade $\varphi^2 \cdot 10^{17}$, and (ii) it lie in the entropic-spinorial window; the identity (68) is the structural bridge between these two requirements.

Remark 7.1 (Phasor identity). *The elementary phase step $\delta\phi = \pi\alpha$ of the filter and the angular scale $\theta_{s1} = \pi/(\sigma\varphi)$ of the principal boundary mode satisfy the closed identity*

$$f + \frac{1}{f} = 2 \cos \theta_{s1} = 2 - \theta_{s1}^2 + O(\theta_{s1}^4), \quad f := e^{i\delta\phi/\varphi}. \quad (69)$$

The quantity $f + 1/f$ is the trace of the kernel phasor; the quantity $2 - (f + 1/f) = \theta_{s1}^2$ is the squared deviation of the phasor from exact closure. The two are the same observable: the kernel phasor and the principal angular scale are one structural invariant, read on the two sides of the projection.

The foregoing establishes the algebraic kernel architecture (the dynamical closure is conditional on the Mellin-diagonal lock-in ansatz): the internal branch $\mathcal{B}_{13} = S_5 \oplus T_8$; the operator identity $\Gamma_{163}^2 = -163 \cdot \mathbf{1}$; the internal splitting $\omega_{\text{bal}}^{(\pm)} = 0.5284 \pm 0.0028$; the Hilbert structure $\Xi_{\ker}(163) = -i\mathcal{H}[\Psi_{\text{boundary}}]$; the two-point structure at $\tau = i$ and τ_{163} ; the Rogers–Ramanujan closure $\varphi^2 = 2 + \mathcal{R}_{\text{QGT}}$; the dynamical and analytic closure theorems with their equivalence (Theorems 7.3–7.5); the Noether-norm derivation of λ_{lock} (Proposition 7.6); the phasor identity (Remark 7.1); and the CM evaluation of H at both special points (Proposition 4.4).

Remark 7.2 (The filter and complex multiplication). *The Pincherle filter $H(z) = P_4(z)/(1-\alpha z)$ is a rational function on $\mathbb{P}^1$. Its arithmetic content is not captured by the notion of modular weight (which applies to modular forms, not rational functions), but by its evaluations at CM points of $\mathbb{Q}(\sqrt{-163})$. At $\tau = i$, the value $H(e^{-2\pi}) = \sigma_0(1 + O(e^{-2\pi}))$ is governed by the golden structure of the lock-in via $\mathcal{R}_{\text{QGT}} = 1/\varphi$. At τ_{163} , the value $H(e^{2\pi i \tau_{163}}) = \sigma_0(1 + O(e^{-\pi\sqrt{163}}))$ collapses to the pure zero mode σ_0 with exponentially small correction, by the transcendence of $e^{\pi\sqrt{163}}$. Both evaluations are proved in this paper. The CM structure of the filter is complete.*

8 Physical and ontological consequences

The hidden sector is not a postulate

Every physical theory that goes beyond the Standard Model must introduce a hidden sector. Supersymmetry doubles the particle content by postulate. Extra-dimensional theories engineer a compact manifold by choice. Dark matter is introduced as a new field with new parameters. In all cases the hidden sector is added to the theory.

$\Xi_{\ker}(163)$ is not added. It is what remains when Π acts. The hidden sector of the QGT is not a postulate but a structural consequence of $\ker \Pi \neq \{0\}$: the bulk has more configurations than the boundary can see, and the configurations it cannot see organise themselves at the domain walls into a Majorana spinor with internal dimension 26 and arithmetic index 163.

The ontological statement is precise:

$$\text{the hidden sector exists because } \ker \Pi \neq \{0\}. \quad (70)$$

Its dimension (26) and its index (163) are not free parameters. They are determined by the rank of Π ($r = 5$), the number of observable boson channels ($|\mathcal{C}_{\text{obs}}| = 3$), and the class-number-1 structure of the imaginary quadratic field $\mathbb{Q}(\sqrt{-163})$. Nothing is inserted from outside.

The kernel is the root of the constants

The boundary reads the lock-in tensor (ξ, η) in two ways [3]. The longitudinal channel ξ gives gravity and the projection residue $\sigma = \alpha^{-1} = 137$. The transverse channel η gives electromagnetism and the electroweak mixing $\sin^2 \theta_W = 3/13$.

The kernel spinor $\Xi_{\ker}(163)$ lives between the two readings. It is the internal object that the domain walls maintain before the projection separates the two channels. Its arithmetic index $163 = 137 + 2 \times 13$ encodes in a single Heegner number the full EM/EW content of the boundary: 137 from the EM residue, 13 from the EW denominator, doubled for the two chiral branches. The boundary sees two constants; the kernel holds one number.

This is a new level of understanding of the fundamental constants. The question of why $\alpha^{-1} \approx 137$ is established in [3]. The question of why $\sin^2 \theta_W = 3/13$ follows from the same rank-5 tensor structure. The deeper question — why these two numbers are related — is answered by the kernel: $163 = \alpha^{-1} + \dim K_{163} = 137 + 26$, and the same integer that is the EW denominator is the internal dimension of each chiral branch. The boundary sees two constants; the kernel holds one number.

What the domain walls emit

If the kernel spinor $\Xi_{\text{ker}}(163)$ is physically real, it leaves traces on the boundary even though it is not itself observable. The predicted signatures are:

(S1) Internal doublet splitting. Every boundary mode at frequency ω_j acquires an internal doublet partner at $\omega_j \pm \varepsilon_{163}$, with splitting $2\varepsilon_{163} \approx 5.6 \times 10^{-3}$. For the balance mode: $\omega_{\text{bal}}^{(\pm)} = 0.5284 \pm 0.0028$. This is in principle distinguishable from the balance mode alone at AFM resolution [5].

(S2) Modular residue in the PSD and CMB compatibility. The residue $\varepsilon_{163} \approx 2.800 \times 10^{-3}$ falls in the entropic-spinorial window:

$$\frac{e}{10^3} < \varepsilon_{163}, \delta_{163} < \frac{2\sqrt{2}}{10^3}.$$

A PSD measurement that resolves structure at the 10^{-3} level would see a splitting that is neither purely entropic nor purely spinorial — it lives strictly between the two boundary registers.

The same window contains the observed CMB temperature:

$$\frac{e}{10^3} < \frac{T_{\text{CMB}}}{10^3 \text{ K}} \approx 2.7255 \times 10^{-3} < \frac{2\sqrt{2}}{10^3}.$$

This common containment is a Level-B structural observation. In the present paper $(\delta_{163}, \varepsilon_{163})$ are the kernel arithmetic residuals of the modular cycle $\sqrt{5} \rightarrow \mathcal{R}_{\text{QGT}} \rightarrow 163$, set by $\pi \notin \mathbb{Q}(\sqrt{5})$; the thermal derivation of T_{CMB} and its harmonic realisation on the CMB sky are downstream of the present kernel result and are not treated in this paper. The window $(e/10^3, 2\sqrt{2}/10^3)$ is a structural band, bounded below by the entropic projection scale e [4] and above by the spinorial breathing attractor $\sqrt{B^*} = 2\sqrt{2}$ (established in [2, 3]). It is recorded as a possible cosmological readout, not a derived constraint.

Three quantities live in the same window: δ_{163} , ε_{163} , and $T_{\text{CMB}}/10^3$. They are not three phenomena that happen to be numerically close. They are three readings of the same fact: the projection does not close exactly, and that non-closure has an arithmetic measure (the modular residual), a thermal measure (the CMB temperature), and a phase measure (the residue δ_{163} in the logarithmic distance between the Ramanujan seed and the golden cascade). $\alpha^{-1} = 137$ and $163 = 137 + 26$ are kernel results; the candidate thermal floor $T_0 = \pi/\ln \pi$ (downstream of this kernel result, not derived here) is the compatible thermal readout of the same non-closure.

Experimental consequence. A PSD measurement at the 10^{-3} level on a system realising the QGT boundary ($k = 2$ lock-in, $n = 5$) should show: (i) the kernel splitting $2\varepsilon_{163} \approx 5.6 \times 10^{-3}$ as a doublet partner of every boundary mode coupled to $\ker \Pi_{\text{loc}}$ within the QGT kernel sector; (ii) a spectral floor $S(f) \propto 1/f$ at $n = 0$, with the characteristic cutoff $f_{\text{min}} = 1$ in natural units of the Majorana 4π cycle; (iii) a $1/f$ thermal floor whose candidate temperature $T_0 = \pi/\ln \pi$ is downstream of this kernel result (not derived here). The modular residual window is the kernel arithmetic fingerprint; the CMB temperature is a compatible thermal readout of the same $n = 0$ floor.

(S2b) CMB anisotropies as indirect observable of the kernel structure. The blackbody temperature T_{CMB} characterises the floor $n = 0$ as a whole. The CMB *anisotropies* $\Delta T/T \sim 10^{-5}$ carry finer information: they are the boundary projection of the fluctuation spectrum of $\xi\eta = \alpha^{-1}$ along the decoherence cascade $n = 5 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$.

Each mode $n \in \mathcal{N}_{\text{adm}}$ extinguishes at a distinct scale and leaves a spectral imprint on the boundary before going dark. The cascade is not uniform: each level n has a characteristic decoherence rate proportional to $\Gamma_\tau \cdot \pi/n$ (the phase window π/n of the n -th admissible filter, weighted by the temporal connection $\Gamma_\tau = \varphi/\sqrt{5}$; the $1/f$ floor as the $n \rightarrow 0$ limit is established in [3]), producing a structured sequence of spectral signatures. The angular power spectrum C_ℓ of CMB anisotropies is, structurally, the boundary readout of this sequence.

Table 1: CMB observables and their structural identification from the kernel spinor $\Xi_{\text{ker}}(163)$.

CMB observable	QGT structural identification
Temperature T_{CMB}	Compatible thermal readout of the decoherent floor $n = 0$; candidate $T_0 = \pi / \ln \pi$, downstream of this result (not derived here).
Window $(e/10^3, 2\sqrt{2}/10^3)$	Entropic-spinorial band of the $n = 0$ boundary sector: bounded below by the entropic scale e of one oscillation and above by the spinorial breathing attractor $2\sqrt{2} = \sqrt{B^*}$.
Anisotropies $\Delta T/T \sim 10^{-5}$	Boundary projection of the cascade $n = 5 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$: each admissible mode extinguishes at rate $\Gamma_\tau \cdot \pi/n$ and leaves a spectral imprint.
Angular spectrum C_ℓ	Projection of the $\dim \mathcal{B}_{13}$ internal kernel modes onto spherical harmonics Y_ℓ^m ; the acoustic peak architecture follows from $[T, x]_{\mathcal{B}_{13}}$ and the dominance of $k = 1$ inside the pole band. Quantitative harmonic realisation: downstream, not treated here.

The connection to the kernel is through the Hilbert decomposition:

$$\Xi_{\text{ker}}(163) = -i \mathcal{H}[\Psi_{\text{boundary}}].$$

The 13 internal modes of $\mathcal{B}_{13} = S_5 \oplus T_8$ are in permanent quadrature with the boundary (ξ and η are phase-shifted by $\pi/2$ by construction). Every mode of the kernel leaves a signature on the boundary. Therefore:

$$\underbrace{\frac{\Delta T}{T}}_{\text{CMB anisotropies}} = \underbrace{\frac{\delta(\xi\eta)}{\xi\eta}}_{\text{boundary fluctuations}} \iff \underbrace{\mathcal{B}_{13}}_{\text{13 kernel modes}}. \quad (71)$$

The CMB anisotropies are an indirect observable of the kernel mode structure: they are the boundary reading of what lives inside $\ker \Pi$.

With $\Psi_\star$ constructed explicitly in Appendix C, the architecture on which the quantitative harmonic realisation of C_ℓ rests is now in place. A first structural estimate of the temperature spectrum reproduces the Planck 2018 peak architecture at RMS $\sim 5\%$. The full harmonic model of D_ℓ^{TT} , D_ℓ^{EE} , D_ℓ^{TE} from the palindromic equation of the rank-5 projection, together with its comparison against Planck 2018 and its relation to ΛCDM , is downstream of the present kernel result and is not treated in this paper.

Here $T_{\text{CMB}}/10^3$ is recorded only as a Level-B compatible readout of the same entropic-spinorial window as the kernel residues; the thermal and harmonic realisations are downstream of this kernel result and are not part of this paper.

(S3) Non-invertibility witness. Two boundary states that differ only by a component of $\Xi_{\text{ker}}(163)$ would be indistinguishable on the boundary: $\Pi\Xi_1 = \Pi\Xi_2$ even though $\Xi_1 \neq \Xi_2$. This is the direct physical realisation of $\ker \Pi \neq \{0\}$: the projection is non-invertible, and the boundary cannot distinguish two bulk configurations that differ in the kernel sector. The experimental fingerprint is the saturation of boundary information at rank 5, regardless of the experimental resolution (Structural Constraints [5]).

(S4) Gravitational entanglement without EM signal. The kernel spinor can carry the bulk invariant $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$ without coupling to the EM channel. Two systems that share a kernel component would exhibit non-classical correlations (EPR=ER in the temporal fibre sense from the rank-5 decoherence cascade) without a classical electromagnetic mediator. The Marletto–Vedral gravitational entanglement protocol would detect the invariant $\mathcal{K}_\tau$ as a non-classical correlation; the absence of an EM signal would signal a kernel origin.

On the Ramanujan–Heegner grammar

The modular layer of the QGT kernel is not a decoration. It is the arithmetic structure of what the projection cannot carry, and it is structurally necessary.

Section B established the central identity: the bulk Majorana spinor carries $\sqrt{5}$ as Clifford spectral curvature; the kernel spinor carries $\mathcal{R}_{\text{QGT}} = 1/\varphi$ as modular signature at the lock-in; and $\varphi + \mathcal{R}_{\text{QGT}} = \sqrt{5}$ is not an observation but the closure relation of $\mathbb{Q}(\sqrt{5})$ (Remark B.1). The two spinors are two readings of the same φ .

The Heegner index 163 enters through (F4): the formula $e^{\pi\sqrt{163}} \approx (2 + \mathcal{R}_{\text{QGT}}) \cdot 10^{17}$ places the maximum arithmetic depth and the kernel’s modular signature in the same expression, with residual $\delta_{163} \in (e/10^3, 2\sqrt{2}/10^3)$ set by $\pi \notin \mathbb{Q}(\sqrt{5})$. The present paper provides the structural explanation for why the kernel admits a modular description: it lives at the CM points of imaginary quadratic fields, accessible via analytic continuation of the Pincherle filter into the upper half-plane (Section 4). $\mathcal{R}_{\text{QGT}} = 1/\varphi$ is the modular value at modulus 5 (the lock-in). $j(\tau_{163}) = -640320^3$ is the value at the maximum Heegner discriminant (the kernel arithmetic index). These are not coincidences seen from outside the theory. They are the natural algebraic language for describing the kernel, once it has been given a precise physical address.

The arithmetic of imaginary quadratic fields — class number 1, Gaussian primes, CM theory — is the algebraic structure underlying the kernel description, parallel to the boundary. In this interpretation, Ramanujan’s CM identities occupy this structure directly, at the $n = 0$ mode where the rank-5 filter is inactive. The boundary receives the modular values as shadows and verifies them by analysis, without seeing their origin in the kernel. This is the structural content of the Hardy–Ramanujan relationship, read in the light of the Hilbert decomposition: the analytic signal on $S^1(\tau)$ is complete only when both the real part (boundary, Hardy) and the imaginary part (kernel, Ramanujan) are present. The programme has now given both a precise address.

Status of the principal results

For transparency we state explicitly, for each principal result of this paper, its epistemic status: *proved* (following from the stated hypotheses), *conditional* (requiring the Mellin-diagonal lock-in ansatz), *structural correspondence* (an exact relation between objects fixed independently, hence not a prediction), or *interpretive* (an assignment of meaning, made without claim of proof).

9 Conclusions

The central result. The QGT kernel is not structurally empty. At the domain walls $DW_{1,2}$, the part of the bulk Majorana spinor that the projection cannot transmit organises itself into a precise internal structure. The filter $H(z) = P_4(z)/(1 - \alpha z)$ is a meromorphic function on the Riemann sphere $\mathbb{P}^1$. Its Hardy–Hilbert decomposition gives:

$$\Xi_{\text{ker}}(163) = H \cdot C\bar{\Psi}_{\text{boundary}} = -i \mathcal{H}[\Psi_{\text{boundary}}], \quad (72)$$

the kernel spinor is the Hilbert transform of the boundary spinor. The internal branch $\mathcal{B}_{13} = S_5 \oplus T_8$ has dimension $13 = 5 + 8$: five from the polynomial degree of P_4 and eight from the tensor degrees of the pole at the lock-in. The full soliton space is $\tilde{H} = \text{span}\{\sigma\} \oplus \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$, $\dim = 27 = 1 + 13 + 13$.

The Riemann sphere and the depth identity. The equator $|z| = 1$ is the boundary; the northern hemisphere $|z| < 1$ is the kernel; the north pole $z = 0$ carries σ_0 ; the south pole $z = \infty$ carries $\sigma = 137$. The conformal map $z = e^{2\pi i\tau}$ translates the Fibonacci language of the boundary into the modular language of the kernel: same function, two sides of the equator.

The depth of the maximum Heegner point satisfies:

$$\text{Im}(\tau_{163}) = 2\pi + \frac{1}{2\kappa^2} + O(10^{-4}), \quad \kappa^2 = 5, \quad (73)$$

Result	Status
Analytic completion $\Xi_{\text{ker}} = -i \mathcal{H}_{\text{Cay}}[\Psi_{\text{bdy}}]$ (Hardy–Hilbert)	<i>Proved</i> (within holomorphy of Ψ_{bdy})
Phasor/spinor partition S_5/T_8 via $N_d^{(2)}$	<i>Proved</i> (set-theoretic, + Baker–Stark)
$\dim \mathcal{B}_{13} = 13$, $\dim K_{163} = 26$	<i>Proved</i> (dimension count)
$\Gamma_{163}^2 = -163 \mathbf{1}_{26}$	<i>Proved</i> , but realised by construction
Noether norm $\lambda_{\text{lock}} = \varphi^2/\sqrt{5}$	<i>Proved</i> (variational)
Dynamical closure $L_*(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$	<i>Conditional</i> on the Mellin-diagonal ansatz
Index $163 = 137 + 26 = \sigma + 2 \dim \mathcal{B}_{13}$	<i>Structural correspondence</i>
mod-13 congruences ($137 \equiv 163 \equiv 7$)	<i>Structural correspondence</i>
Thermal readout band	<i>Structural correspondence</i> ; downstream, not treated here
Ramanujan–Heegner grammar	<i>Interpretive</i>

Table 2: Epistemic status of the principal statements. “Proved” follows from the stated hypotheses; “conditional” requires the Mellin-diagonal lock-in ansatz; “structural correspondence” is an exact relation between independently fixed objects and is therefore not a prediction; “interpretive” is an assignment of meaning made without claim of proof. The relations $163 = 137 + 26$ and the mod-13 congruences are exact and internally motivated, but the summands are fixed independently (by Baker–Stark, by the kernel dimension count, and by the boundary coupling); they are recorded as correspondences, not derivations.

one full boundary perimeter (2π) plus one minimal cascade step ($1/10$). The modular residue $\varepsilon_{163} = \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17}) \approx 2.800 \times 10^{-3}$ is the phase gap between the geometric depth and the golden cascade, rooted in $\pi \notin \mathbb{Q}(\sqrt{5})$ — the same transcendence that drives the arrow of time. The incompleteness of the arithmetic cascade is the physics.

Three structural consequences. The Hardy–Hilbert decomposition of $H(z)$ has three direct physical readings. The kernel spinor $\Xi_{\text{ker}}(163) = -i\mathcal{H}[\Psi_{\text{boundary}}]$ means: (i) the hidden sector is the anti-causal component of the same filter that generates the boundary — it is not added, it is required; (ii) the Hardy analytic signal $f_H = q + ip$ carries the boundary/kernel quadrature with conserved instantaneous frequency $\dot{\phi} = \Gamma_\tau = \varphi/\sqrt{5}$, while the chiral doublet (ξ, η) carries the hyperbolic lock-in $\xi\eta = \alpha^{-1}$ — with ξ (longitudinal, gravity) and η (transverse, EM) the two projective readouts; (iii) the non-closure $\Delta\phi = 2\pi\Gamma_\tau \notin 2\pi\mathbb{Q}$ is forbidden from vanishing by $\pi \notin \mathbb{Q}(\sqrt{5})$ (Lindemann–Weierstrass), and this arithmetic obstruction is the structural origin of the temporal ordering established in [4]. Physical and interpretive implications are discussed in the Epilogue.

Five interlocking facts. (1) 163 largest Heegner number (Baker–Stark). (2) $\sigma = 137$ Gaussian prime in $\mathcal{O}_{163}$. (3) $13 = 2r + |\mathcal{C}_{\text{obs}}| = 5 + 8 = \dim \mathcal{B}_{13} = 1/\sin^2 \theta_W \cdot 3$. (4) $\mathcal{R}_{\text{QGT}} = 1/\varphi$, $\varphi + R = \sqrt{5}$. (5) $\text{Im}(\tau_{163}) = 2\pi + 1/(2\kappa^2)$. Remove any one: the arithmetic coherence collapses. Together: 163 is not a candidate — it is the unique discriminant forced by the rank-5 Majorana projection with three observable boson channels.

The conditional closures. The dynamical invariance $L_*(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$ (Theorem 7.3) and the analytic closure $P^-(H(e^{2\pi i\tau_{163}})\Xi_*) \in \mathcal{B}_{13}$ (Theorem 7.4) are both established conditionally, under the Mellin-diagonal lock-in ansatz (Appendix C). The Noether-norm derivation of the lock-in coupling (Proposition 7.6) fixes $\lambda_{\text{lock}} = \varphi^2/\sqrt{5}$ from the rank-5 projection geometry alone. The thermal-damping identity $\ell_T = 163 + \lambda_{\text{lock}}^2$ (Remark C.2) closes the chain from arithmetic kernel index to observable sky scale on a single equation. The quantitative harmonic realisation of the CMB spectrum is downstream of this kernel result and is not treated in this paper.

The cycle. The structure closes on itself:

$$\begin{aligned}
\sqrt{5} &\xrightarrow{\Pi} \varphi \text{ (boundary)} + \underbrace{\mathcal{R}_{\text{QGT}}}_{=1/\varphi} \text{ (kernel)} \\
\mathcal{R}_{\text{QGT}} &\xrightarrow{\varphi^2=2+R} e^{\pi\sqrt{163}} \approx (2+R) \cdot 10^{17} \\
\delta_{163} &\in \left(\frac{e}{10^3}, \frac{2\sqrt{2}}{10^3} \right) \xrightarrow{\pi \notin \mathbb{Q}(\sqrt{5})} \text{time} \\
\frac{T_{\text{CMB}}}{10^3 \text{ K}} &\in \left(\frac{e}{10^3}, \frac{2\sqrt{2}}{10^3} \right) \xrightarrow{1/f \text{ floor}, \Gamma_\tau^{(0)}=4\pi} T_0 = \frac{\pi}{\ln \pi}
\end{aligned} \tag{74}$$

The projection separates φ from $1/\varphi$. The Hilbert transform exchanges them. The CM theory of $\mathbb{Q}(\sqrt{-163})$ collects them into 163. The residual δ_{163} measures how far the cycle is from closing exactly. That non-closure — forbidden by Lindemann–Weierstrass — is time.

Epilogue: what the structure has shown

The programme began with a single structural observation: the observable boundary cannot determine its own admissible algebra. Something outside it must select what is observable in principle. That something is the projection Π — non-invertible, finite-rank, irreversible.

From that one observation, without further assumptions, the programme derived: the five admissible modes, the fine-structure constant, the electroweak mixing angle, the Standard Model gauge group, the arrow of time, the Born rule, and the breathing attractor of boundary dynamics [1, 2, 3, 4]. Each result came from the structure itself. Nothing was inserted from outside.

This paper has derived one more layer of the same structure: the kernel — what Π cannot transmit — is not empty, not arbitrary, and not separate from the boundary. It is the boundary's own imaginary part. The same filter $H(z)$ that generates the observable physics on the equator of $\mathbb{P}^1$ generates, in its northern hemisphere, the organised hidden structure $\mathcal{B}_{13} = S_5 \oplus T_8$ with arithmetic index 163. The three generating constants π , e , $\alpha^{(B)}$ — boundary closure, entropic scale, observable residue — appear each at their natural cascade level in the depth $\text{Im}(\tau_{163})$ of the maximum Heegner point. The structure wrote this; we read it.

The programme has traced a chain of four arrows. Each arrow is physically necessary, not arithmetically convenient.

$$\begin{aligned}
\sqrt{5} &\xrightarrow{\text{spectral curvature}} \alpha^{-1} = 137 \\
&\xrightarrow{\text{topological index}} 163 = 137 + 26 \\
&\xrightarrow{\text{modular readout}} \mathcal{R}_{\text{QGT}} = 1/\varphi \\
&\xrightarrow{\text{thermal floor}} T_0 = \pi / \ln \pi.
\end{aligned}$$

First arrow: spectral curvature $\rightarrow$ irreducible residue. $\sqrt{5}$ is not chosen. It is the dominant eigenvalue of the Pincherle dilatation $D_\tau(\varphi^2) = 2\varphi^2$ on the compact temporal fibre $S^1(\tau)$: the curvature the bulk acquires when a structure with golden autosimilarity $\varphi^2 = \varphi + 1$ projects onto itself. This curvature determines how much of the bulk can be resolved at the boundary: rank 5, five admissible modes, five independent angular channels. The residue left on the boundary after all five channels are exhausted is $\sigma = \lfloor \alpha^{-1} \rfloor = 137$ — a Gaussian prime in $\mathcal{O}_{163}$, irreducible by number-theoretic necessity. The fine-structure constant is the arithmetic remainder of a geometric act. It cannot be different, because the curvature $\sqrt{5}$ cannot be different given φ .

Second arrow: irreducible residue $\rightarrow$ topological index of the kernel. $163 = 137 + 26$ is not a sum. It is a partition forced by three simultaneous constraints: the projection residue ($\sigma = 137$, established in Appendix D), the kernel internal dimension ($26 = 2 \times \dim \mathcal{B}_{13}$, fixed by the filter $H(z)$ of this paper), and the requirement that 163 be the largest Heegner discriminant with class number 1 (Baker–Stark theorem: no choice, only 163). The physical meaning: 163 is the address of the maximum depth at which the structure that the boundary cannot observe — the kernel — organises itself. At that depth and no other, the projection residue ($\sigma = 137$, what the boundary keeps) and the kernel dimension (26, what the boundary loses) partition the unique discriminant that allows both to be prime and class-1 simultaneously. The boundary is stable (σ prime, cannot decompose further); the kernel is organised (26 dimensional, not random); 163 is the number that makes both possible at once.

Third arrow: topological index $\rightarrow$ modular readout. The filter $H(e^{2\pi i\tau})$ evaluated at the lock-in point $\tau = i$ reads the golden return $\mathcal{R}_{\text{QGT}} = 1/\varphi$ — the Fibonacci pole and first singular value of the rank-5 projection, distinct from the Rogers–Ramanujan value at $q = e^{-2\pi}$ (≈ 0.284), a structure governed by complex multiplication theory on $\mathbb{Q}(\sqrt{-163})$. This is not inserted. It follows from where the filter lives. At the point of maximum arithmetic depth, the modular value $\mathcal{R}_{\text{QGT}} = 1/\varphi$ involves the same φ that generates the bulk spectrum $\sqrt{5} = 2\varphi - 1$, now appearing in the kernel as $1/\varphi$ under the field automorphism of $\mathbb{Q}(\sqrt{5})$. The projection separates φ from $1/\varphi$. The Hilbert transform exchanges them. The CM theory of $\mathbb{Q}(\sqrt{-163})$ collects them into 163 — the unique point where separation and reunion become arithmetic.

$$\varphi + \mathcal{R}_{\text{QGT}} = \sqrt{5} :$$

the bulk and the kernel are not two objects. They are two projections of the same φ , forced apart by Π and reunited by 163.

Fourth arrow: non-closure $\rightarrow$ thermal floor. The non-closure $\Delta\phi = 2\pi\Gamma_\tau \notin 2\pi\mathbb{Q}$ is not a defect — it is the engine. $\pi \notin \mathbb{Q}(\sqrt{5})$ (Lindemann–Weierstrass) prevents the arithmetic cycle from closing exactly. The residue δ_{163} measures this gap; it lives in the window $(e/10^3, 2\sqrt{2}/10^3)$, bounded below by the entropic scale e (what one oscillation costs in information) and above by the spinorial scale $2\sqrt{2} = \sqrt{B^*}$ (what the boundary can sustain before losing coherence). At the floor $n = 0$, when every admissible mode has extinguished, the Majorana cycle 4π is freed of all modal compression: $\Gamma_\tau^{(0)} = 4\pi$. One decoherent oscillation carries energy π and produces entropy $\ln \pi$. Downstream of this kernel result, the same non-closure is proposed to yield a candidate thermal floor

$$T_0 = \frac{\pi}{\ln \pi} \approx 2.7444 \text{ K},$$

compatible, after the spinorial correction $(\pi/\ln \pi)(1 - 1/(\sqrt{2} \cdot 100)) \approx 2.724991 \text{ K}$, with the measured T_{CMB} ; that derivation is not part of this paper. In the present paper $T_{\text{CMB}}/10^3$ is only a compatible Level-B readout of the same entropic–spinorial window as δ_{163} : the arithmetic residue and the thermal residue are two readings of the same irreducible non-closure.

The CMB anisotropies $\Delta T/T \sim 10^{-5}$ carry one further layer: they are the boundary projection of the decoherence cascade $n = 5 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$, each mode extinguishing at its characteristic rate $\Gamma_\tau \cdot \pi/n$ and leaving a spectral imprint before going dark. The angular power spectrum C_ℓ is the boundary readout of the 13 modes of $\mathcal{B}_{13}$, projected onto spherical harmonics. Downstream of this kernel result, the CMB temperature is proposed to arise as a candidate thermal floor compatible with the measured value; in the present paper it is only a Level-B compatible readout of the arithmetic residue, not a derived quantity, and the quantitative realisation of C_ℓ is not treated here.

The algebraic kernel architecture is closed; the dynamical closure is conditional on the Mellin-diagonal lock-in ansatz. The explicit seed $\Psi_\star$ (Appendix C) instantiates the construction; the Mellin-diagonal action of $L_\star$ establishes the invariance of $\mathcal{B}_{13}$; the analytic closure at τ_{163} confirms

both from the CM-point side. The lock-in coupling λ_{lock} is derived, not fitted. The identity $\varphi + \mathcal{R}_{\text{QGT}} = \sqrt{5}$ is the arithmetic seal. The CM structure of the filter is proved at both special points.

The following reading is interpretive, not part of the formal argument. The structure derived here admits a unified reading. The projection Π separates two parts of the same bulk object: the boundary spinor Ψ_{boundary} , which is observable, and the kernel spinor $\Xi_{\text{ker}}(163)$, which is not. The two are related by a Hilbert transform — the imaginary part of an analytic signal is not independent of its real part, it is required by it. The non-invertibility of Π means that what enters the kernel does not return to the boundary: this is the structural origin of temporal irreversibility, as established in [4]. The arithmetic obstruction $\pi \notin \mathbb{Q}(\sqrt{5})$ prevents the phase cycle from closing exactly; the residue δ_{163} quantifies this gap; the same irrational π that prevents closure determines the thermal scale $T_0 = \pi / \ln \pi$ of the decoherent floor.

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A The Heegner phasor algebra and the quadratic partition

The numbers 5, 8, 13, 19, 43, 67, 137, 163 that appear in the kernel structure are not parameters. They are readouts of a single algebraic object: the quadratic norm forms of imaginary quadratic fields of class number one. This section derives the partition $\mathcal{B}_{13} = S_5 \oplus T_8$ from first principles of algebraic number theory, establishes the origin of the filter pole α_{Π}^{-1} as a Fibonacci quadratic norm, and reads the domain wall algebra from the mod-13 residue structure.

A.1 Nine fields, nine norm forms

The class-number-one theorem of Baker and Stark [11, 12] asserts that there are exactly nine imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$ with class number $h(K_d) = 1$. Class number one means unique factorisation in the ring of integers $\mathcal{O}_{K_d}$: every ideal factors uniquely, and every element's norm is represented by a single quadratic form with no fragmentation. This arithmetic rigidity is what makes a field's structure *stable as a memory mode*.

The ring of integers of $\mathbb{Q}(\sqrt{-d})$ is

$$\mathcal{O}_{K_d} = \begin{cases} \mathbb{Z}\left[\frac{1+\sqrt{-d}}{2}\right] & d \equiv 3 \pmod{4}, \\ \mathbb{Z}[\sqrt{-d}] & d \equiv 1, 2 \pmod{4}, \end{cases} \quad (75)$$

with quadratic norm $N_d(z) = z\bar{z} \in \mathbb{Z}_{\geq 0}$ for $z \in \mathcal{O}_{K_d}$.

d	ω_d (generator)	Norm form $N_d(a, b)$	$N_d^{(2)}$	Role
1	i	$a^2 + b^2$	2	H_{bdy}
2	$\sqrt{-2}$	$a^2 + 2b^2$	2	H_{bdy}
3	$\frac{1+\sqrt{-3}}{2}$	$a^2 + ab + b^2$	3	H_{bdy}
7	$\frac{1+\sqrt{-7}}{2}$	$a^2 + ab + 2b^2$	2	H_{bdy}
11	$\frac{1+\sqrt{-11}}{2}$	$a^2 + ab + 3b^2$	3	H_{bdy}
19	$\frac{1+\sqrt{-19}}{2}$	$a^2 + ab + 5b^2$	5	H_{ker} , gen. 1
43	$\frac{1+\sqrt{-43}}{2}$	$a^2 + ab + 11b^2$	11	H_{ker} , gen. 2
67	$\frac{1+\sqrt{-67}}{2}$	$a^2 + ab + 17b^2$	17	H_{ker} , gen. 3
163	$\frac{1+\sqrt{-163}}{2}$	$a^2 + ab + 41b^2$	41	ribbon closure

Here $N_d^{(2)}$ is the smallest norm value attained for $(a, b) \neq (1, 0)$ — the *second distinct norm* after the minimal value $N_d(1, 0) = 1$. The partition $H = H_{\text{bdy}} \cup H_{\text{ker}} \cup \{163\}$ is determined entirely by the spectral sequence of the norm form — not by the discriminant values themselves.

A.2 The partition via $N_d^{(2)}$: phasor vs spinorial depth

Before the algebra, the geometry.

Every class-number-one field $\mathbb{Q}(\sqrt{-d})$ defines a phasor $\rho_d \in U(1)$: a unit complex number built from the minimal non-unit element of its ring of integers. The phasor traces a trajectory

on the circle S_χ^1 as the temporal fibre parameter τ advances. The question that determines the partition is simple: *does the trajectory close?*

For the five boundary fields $d \in \{1, 2, 3, 7, 11\}$, the answer is yes. The phasor closes in $m = 2p$ steps, where $p \in \mathcal{N}_{\text{adm}}$ is the corresponding admissible mode. The closure is exact, cyclotomic, and independent of d : the second norm $N_d^{(2)} \in \{2, 3\}$ stays fixed on the two smallest primes regardless of how large d grows. These phasors *live on the circle*: they perceive only angle, not depth. They are the boundary modes of the QGT — what Π carries across to $\mathbb{R}^4$.

For the three kernel fields $d \in \{19, 43, 67\}$, the trajectory does not close on S_χ^1 . The second norm $N_d^{(2)} \in \{5, 11, 17\}$ grows with d as $N_d^{(2)} \approx d/4$ — the phasor acquires spinorial depth, a capacity to perceive distance in the arithmetic of $\mathcal{O}_{K_d}$ that no cyclotomic rotation can supply. To close its holonomy, the phasor must be *lifted*: the Majorana self-conjugation condition $\Xi = C\bar{\Xi}$ maps $s_d \mapsto \chi_d = (s_d, \bar{s}_d)^\top \in S_\chi^2$, taking the phasor from the circle to the two-sphere. Each generation opens one level of spinorial depth that the boundary cannot see.

For $d = 163$, the second norm $N_{163}^{(2)} = 41$ exceeds the saturation index $n_{\text{sat}} = 11$ of the admissible catalogue. No mode in $\mathcal{N}_{\text{adm}}$ can accommodate a spinorial step of size 41. The trajectory cannot close at any level accessible to the projection Π . This is not a failure of a particular lift: it is the statement that the arithmetic of $\mathbb{Q}(\sqrt{-163})$ lies beyond the reach of Π entirely. The field $d = 163$ is therefore not a generation of the kernel — it is the arithmetic witness that the list of generations is complete. Baker and Stark proved that no further class-number-one field exists beyond $d = 163$ [11, 12]; the QGT reads this as the closure of the admissible catalogue.

The second norm $N_d^{(2)}$ is the discriminating quantity:

Sector	S_5 (boundary)					T_8 (kernel)			$\{163\}$
d	1	2	3	7	11	19	43	67	163
$N_d^{(2)}$	2	2	3	2	3	5	11	17	41

The S_5 fields ($d \in \{1, 2, 3, 7, 11\}$) have $N_d^{(2)} \in \{2, 3\}$ — fixed on the two smallest primes regardless of d . The T_8 fields ($d \in \{19, 43, 67\}$) have $N_d^{(2)} \in \{5, 11, 17\}$, growing as $N_d^{(2)} \approx d/4$ with attractor $2^2 = 4$.

This is not a quantitative distinction — it is a qualitative one.

The S_5 fields are *pure phasors*: their second norm stays on $\{2, 3\}$ because each $\mathbb{Q}(\sqrt{-d})$ for $d \in H_{\text{bdy}}$ embeds into a cyclotomic field

$$\mathbb{Q}(\sqrt{-d}) \hookrightarrow \mathbb{Q}(\zeta_m), \quad m \in \{4, 8, 6, 14, 22\} = \{2^2, 2^3, 2 \cdot 3, 2 \cdot 7, 2 \cdot 11\}, \quad (76)$$

with holonomy of order $m = 2p$ or $m = 2^k$, $p \in \mathcal{N}_{\text{adm}}$. A pure phasor lives on S_χ^1 : it perceives only angle (rotation order), not depth (spinorial distance). The norm $N_d^{(2)} \in \{2, 3\}$ is the smallest non-trivial element of the cyclotomic rotation group, independent of d .

The T_8 fields are *spinorial*: $N_d^{(2)} \sim d/4$ grows with d because these fields cannot close their holonomy at the cyclotomic level. They require a Majorana lift from S_χ^1 to the spinorial two-sphere S_χ^2 . The phasor perceives depth; the norm measures it.

Proposition A.1 (Algebraic derivation of the S_5/T_8 partition). *The internal branch $\mathcal{B}_{13} = S_5 \oplus T_8$ is uniquely determined by the class-number-one norm structure: $S_5 = \{d \in H : N_d^{(2)} \in \{2, 3\}\} = \{1, 2, 3, 7, 11\}$ (cyclotomic holonomy, pure phasors); $T_8 = \{d \in H : N_d^{(2)} \in \{5, 11, 17\}\} = \{19, 43, 67\}$ (spinorial holonomy, Majorana-liftable kernel branch). The field $d = 163$ has $N_{163}^{(2)} = 41$ and falls outside both sectors: it closes the Baker–Stark list without contributing a fermionic generation.*

Proof. The rank-5 projection admits modes $n \leq n_{\text{sat}} = 11$ [3]. A field's $N_d^{(2)}$ measures the minimal spinorial step its phasor requires to distinguish from a unit. For $N_d^{(2)} \in \{2, 3\}$ (the fields $d \in \{1, 2, 3, 7, 11\}$): the step falls within the polynomial sector $P_4(z)$, whose five logarithmic modes span S_5 . The cyclotomic embedding (76) holds precisely for these five fields; their phasors close within S_χ^1 . For $N_d^{(2)} \in \{5, 11, 17\}$ (the fields $d \in \{19, 43, 67\}$): the step exceeds the polynomial capacity but lies within the Majorana-liftable kernel branch, with 17 the outer generation beyond $\mathcal{N}_{\text{adm}}$. Their phasors require the Majorana lift $s_d \mapsto \chi_d = (s_d, \bar{s}_d)^\top \in S_\chi^2$; they generate T_8 . For $N_{163}^{(2)} = 41$: the step exceeds every value realised by the admissible catalogue; no further spinorial mode can accommodate it. The field $d = 163$ is the arithmetic witness of saturation. $\square$

The ontological reading of this partition is precise. The projection Π is not a map chosen to match observations. It is the unique rank-5 self-similar projection forced by the Fibonacci closure condition $F_r = r$ with $r = 5$ [2]. The admissible catalogue $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ is forced by the same condition. The class-number-one fields of imaginary quadratic type are forced by algebraic number theory — they are what they are, independent of any physical input.

What the QGT discovers is that these two forced structures *fit together exactly*: the fields whose phasors close cyclotomically ($N_d^{(2)} \in \{2, 3\}$) are precisely the fields whose modes are observable at the boundary; the fields whose phasors require a spinorial lift ($N_d^{(2)} \in \{5, 11, 17\}$) are precisely the fields whose modes constitute the internal kernel.

Nothing is adjusted to make this work. The boundary and the kernel are not designed to complement each other. They are the two sectors that the same arithmetic constraint — class number one — distributes across the same projection threshold $r = 5$. The partition $S_5 \oplus T_8$ is not a decomposition chosen after the fact. It is the form that $\ker \Pi$ must take if the admissible catalogue is $\mathcal{N}_{\text{adm}}$ and the class-number-one list is what Baker and Stark proved it to be.

A.3 Explicit phasor construction for $d \in \{19, 43, 67\}$

The Heegner phasor for each kernel field is built from the minimal non-unit element of $\mathcal{O}_{K_d}$, which is the ring generator $\omega_d = (1 + \sqrt{-d})/2$ itself (for $d \equiv 3 \pmod{4}$, as all three satisfy):

Generation 1 — $d = 19$. $N_{19}(\omega_{19}) = (1 + 19)/4 = 5 = N_{19}^{(2)}$.

$$z_{19} = \frac{1 + i\sqrt{19}}{2}, \quad N_{19}(z_{19}) = 5, \quad (77)$$

$$\rho_{19} = \frac{z_{19}}{|z_{19}|} = \frac{1 + i\sqrt{19}}{2\sqrt{5}} \in U(1), \quad (78)$$

$$\chi_{19} = \begin{pmatrix} s_{19}(\tau) \\ \bar{s}_{19}(\tau) \end{pmatrix} \in S_\chi^2, \quad s_{19}(\tau) = \rho_{19} e^{im_{19}\tau}. \quad (79)$$

Condensation eigenvalue $\lambda_{19} \propto N_{19}^{(2)} = 5 = r$: the first kernel generation opens at the projection rank.

Generation 2 — $d = 43$. $N_{43}(\omega_{43}) = (1 + 43)/4 = 11 = n_{\text{sat}}$.

$$z_{43} = \frac{1 + i\sqrt{43}}{2}, \quad N_{43}(z_{43}) = 11, \quad (80)$$

$$\rho_{43} = \frac{1 + i\sqrt{43}}{2\sqrt{11}} \in U(1), \quad \chi_{43} = \begin{pmatrix} s_{43} \\ \bar{s}_{43} \end{pmatrix}. \quad (81)$$

Condensation eigenvalue $\lambda_{43} \propto N_{43}^{(2)} = 11 = n_{\text{sat}}$: the second generation saturates the admissible catalogue.

Generation 3 — $d = 67$. $N_{67}(\omega_{67}) = (1 + 67)/4 = 17$.

$$z_{67} = \frac{1 + i\sqrt{67}}{2}, \quad N_{67}(z_{67}) = 17, \quad (82)$$

$$\rho_{67} = \frac{1 + i\sqrt{67}}{2\sqrt{17}} \in U(1), \quad \chi_{67} = \begin{pmatrix} s_{67} \\ \bar{s}_{67} \end{pmatrix}. \quad (83)$$

Condensation eigenvalue $\lambda_{67} \propto N_{67}^{(2)} = 17$: the third generation has no admissible mode partner ($17 \notin \mathcal{N}_{\text{adm}}$), which is the structural signature that it is the outermost generation.

The three phasors ρ_{19} , ρ_{43} , ρ_{67} generate the three spinorial columns of T_8 . Their condensation eigenvalues $\{5, 11, 17\}$ are the three values of $N_d^{(2)}$ for $d \in H_{\text{ker}}$ — neither input nor fit, but read off the norm form of each field.

A.4 Bridge 4: the filter pole as a Fibonacci quadratic norm

The pole of the Pincherle filter at $z_{\text{pole}} = \alpha_{\Pi}^{-1}$ is not a number chosen to match the fine-structure constant. It is the quadratic norm of the Fibonacci vector $(F_5, F_4)^{\top}$ in the metric of $\mathbb{Z}[\sqrt{5}]$:

$$\alpha_{\Pi}^{-1} = (5 + 3\sqrt{5})^2 = 20\varphi^4 = c_F^{\top} G_{\sqrt{5}} c_F, \quad c_F = \begin{pmatrix} F_5 \\ F_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \quad (84)$$

where $G_{\sqrt{5}} = \begin{pmatrix} 1 & \sqrt{5} \\ \sqrt{5} & 5 \end{pmatrix}$ is the Gram matrix of $\mathbb{Z}[\sqrt{5}]$ in the basis $(1, \sqrt{5})$. The closure condition $F_r = r$ (Fibonacci self-similarity of the projection, $F_5 = 5 = r$) selects the vector $c_F = (F_5, F_4)^{\top} = (5, 3)^{\top}$; the quadratic norm in $\mathbb{Q}(\sqrt{5})$ then gives $(5 + 3\sqrt{5})^2 = 20\varphi^4$.

The integer part is $\sigma = \lfloor \alpha_{\Pi}^{-1} \rfloor = 137$. This is not reverse-engineered from the fine-structure constant: it is the integer part of a Fibonacci quadratic norm, forced by $F_r = r$ and the golden field $\mathbb{Q}(\sqrt{5})$.

The bridge identity that ties the continuous and discrete languages:

$$\varphi + \frac{1}{\varphi} = \sqrt{5}, \quad \text{i.e.} \quad \varphi + \mathcal{R}_{\text{QGT}} = \sqrt{5}, \quad (85)$$

where $\mathcal{R}_{\text{QGT}} := 1/\varphi$ is both the first singular value σ_1 of the rank-5 projection [3] and the arithmetic return of the kernel. It is the QGT golden return — the fixed point of $R = 1/(1 + R)$, equivalently $R^2 + R - 1 = 0$ — and is *distinct* from the standard Rogers–Ramanujan value at $q = e^{-2\pi}$, which by Berndt’s evaluation is $R_{\text{RR}}(e^{-2\pi}) = \sqrt{(5 + \sqrt{5})/2} - \varphi \approx 0.284$; the two coincide only in the limit $q \rightarrow 1$, where the Rogers–Ramanujan continued fraction degenerates to the golden ratio. The boundary reads $\sqrt{5}$ as the Clifford spectral curvature of the bulk; the kernel returns $1/\varphi$ as its modular echo. Projection and kernel are two readings of the same quadratic field $\mathbb{Q}(\sqrt{5})$.

A.5 The discriminant scaling chain

The nine Heegner discriminants are not scattered. They form a chain determined by three structural constants alone: the saturation index $n_{\text{sat}} = 11$, the Majorana breathing quantum $B^* = 8 = 2^3$, and the ribbon dimension $\dim \mathcal{B}_{13} = 13$.

$$\begin{aligned}
n_{\text{sat}} &= 11 && \text{seed: saturation mode of } \mathcal{N}_{\text{adm}} \\
11 + B^* &= 19 && \text{generation 1: one breathing quantum} \\
19 + 3B^* &= 43 && \text{generation 2: three EW traversals} \\
43 + 3B^* &= 67 && \text{generation 3: confirmed period } 3B^* = 24 \\
67 + n_{\text{conf}} \cdot 2r &\approx 137 \approx \alpha_{\Pi}^{-1} && \text{(boundary pole, integer part)} \\
137 + 2 \dim \mathcal{B}_{13} &= 163 && \text{ribbon closure (Baker–Stark)}
\end{aligned} \tag{86}$$

with $B^* = 8 = 2^3$ (Majorana holonomy: $2 \times 4\pi$ spinorial cycle), $n_{\text{conf}} = 7$ (the confined mode of $\mathcal{N}_{\text{adm}}$), $r = 5$ (projection rank), $|\mathcal{C}_{\text{obs}}| = 3$ (observable boson channels), $3B^* = 24$ (period $= |\mathcal{C}_{\text{obs}}| \times B^*$).

The generating steps $\{8, 24, 70, 26\}$ are themselves structural: 8 is the minimal Majorana holonomy; $24 = 3 \times 8$ is the three-channel EW period; the step from generation 3 to the boundary pole is $137 - 67 = 70 = n_{\text{conf}} \times 2r = 7 \times 10$, where the confined mode $n_{\text{conf}} = 7$ and the monodromy norm $2r = 10$ multiply; the final step $26 = 2 \times 13 = 2 \dim \mathcal{B}_{13}$ is the Majorana double cover of the ribbon.

Every discriminant value in the chain is a readout of this structure, not an input to it. The chain also explains the mod-24 congruence: all kernel fields satisfy $d \equiv 3 \pmod{8}$ (norm form with odd middle coefficient, forced by the ring structure), giving combined period $\text{lcm}(8, 3) = 24$, hence $19 \equiv 43 \equiv 67 \equiv 163 \equiv 19 \pmod{24}$.

A.6 The mod-13 algebra at the domain wall

The domain wall $DW_{1,2}$ is a gauge-selective filter. The mod-13 structure of the kernel closure encodes precisely what the filter passes and what it retains.

The three key values satisfy:

$$163 \equiv 7 \pmod{13}, \quad 137 \equiv 7 \pmod{13}, \quad 26 \equiv 0 \pmod{13}. \tag{87}$$

The residue class 7 is the signature of the confined mode $n = 7 \in \mathcal{N}_{\text{adm}}$: $n = 7$ does not generate the Pincherle dilation lattice on the boundary and lives only in $\ker \Pi$. Both the kernel closure 163 and the boundary pole $\sigma = 137$ carry this signature. Their difference $26 = 163 - 137$ is exactly zero modulo $\dim \mathcal{B}_{13} = 13$: it is *invisible at the boundary*.

The shadow ribbon $\mathcal{B}_{13}^{\text{shadow}}$ is the unobservable complement of $\mathcal{B}_{13}$ at the domain wall:

$$\Pi : \mathcal{B}_{13}^{\text{core}} \longrightarrow \mathcal{B}_{13}^{\text{bdy}}, \quad \ker(\Pi|_{\mathcal{B}_{13}}) \cong \mathcal{B}_{13}^{\text{shadow}}, \quad \dim \mathcal{B}_{13}^{\text{shadow}} = 26. \tag{88}$$

Its internal decomposition:

$$26 = 2 \times (5 + 8) = \underbrace{2 \times 5}_{10} + \underbrace{2 \times 8}_{16} = 2 \dim S_5 + 2 \dim T_8. \tag{89}$$

The projection loses exactly one full ribbon. What passes has residue $\equiv 7 \pmod{13}$ (via the boundary pole σ); what stays has residue $\equiv 0 \pmod{13}$ and is invisible to the observable algebra.

This is the complete statement of the domain wall as an algebraic object. The quantitative derivation of what the domain wall *emits* — the edge map $G_{\text{edge}, \mathcal{H}}^{\eta\xi} : S_5 \rightarrow T_8$ that governs transitions between the boundary and kernel sectors, and in particular the $\pi^+ \rightarrow \mu^+ + \nu_\mu$ transition where the pion (boundary, $\in S_5$) decays to the muon (kernel, $\in T_8$) with the neutrino carrying the mod-13 invisible charge — belongs to the matter sector and is deferred to a subsequent paper.

A.7 The analytic signal as sech/tanh decomposition

The connection between the Heegner phasor algebra and the Hardy–Hilbert decomposition of Section 4 is made concrete by the mother solution.

The bulk Majorana field at the lock-in takes the form:

$$\Xi_{\text{bulk}} = \varphi \operatorname{sech}(\Gamma_\tau t) - i\varphi \tanh(\Gamma_\tau t), \quad (90)$$

where $\operatorname{sech}(\Gamma_\tau t) = \operatorname{Re} \Xi_{\text{bulk}}$ is the boundary component Ψ_{bdy} and $\tanh(\Gamma_\tau t) = (-1/\varphi) \operatorname{Im} \Xi_{\text{bulk}}$ is proportional to Ξ_{ker} . The Hilbert–Cayley identity

$$\mathcal{H}_{\text{Cay}}[\operatorname{sech}(\Gamma_\tau t)] = \tanh(\Gamma_\tau t) \quad (91)$$

is the operator realisation of Proposition 4.1 ($\Xi_{\text{ker}} = -i\mathcal{H}[\Psi_{\text{bdy}}]$) in the explicit basis of the mother solution. The conservation law is:

$$\operatorname{sech}^2(\Gamma_\tau t) + \tanh^2(\Gamma_\tau t) = 1, \quad (92)$$

which is the constant-modulus condition $|\Xi_{\text{bulk}}|^2 = \varphi^2$: the total amplitude of the analytic signal is conserved, and the projection Π merely rotates between its real and imaginary parts.

The Heegner phasor algebra now closes on the analytic signal: the phasors $\rho_d = (1 + i\sqrt{d})/(2\sqrt{N_d^{(2)}})$ are the kernel’s arithmetic skeleton at the CM points τ_d , the sech / tanh decomposition is their dynamical realisation on the temporal fibre, and the Hilbert transform is the operator that exchanges the two. Neither can exist without the other: $\operatorname{sech}^2 + \tanh^2 = 1$ is the same identity as $\varphi + 1/\varphi = \sqrt{5}$, read as an energy conservation law instead of an algebraic closure.

Sections 2 and 3 established the location and the internal structure of the kernel. This section gives the kernel a name and an operator. The kernel spinor $\Xi_{\text{ker}}(163)$ is not a new field introduced to account for missing physics. It is what the bulk Majorana spinor $\Xi = C\bar{\Xi}$ becomes when Π acts and cannot transmit it. It is already there. The only question is what it looks like.

Definition and Majorana structure

The kernel spinor is the component of the full bulk Majorana field $\Xi = C\bar{\Xi}$ that lives in the local kernel at the domain walls:

$$\begin{aligned} \Xi_{\text{ker}}(163) &:= (\mathbf{1} - \Pi)\Xi|_{\text{ker } \Pi^{\text{loc}}}, \\ \Xi_{\text{ker}}(163) &= C \overline{\Xi_{\text{ker}}(163)}. \end{aligned} \quad (93)$$

The Majorana condition is inherited: since $\Xi = C\bar{\Xi}$ holds for the full bulk field, and Π commutes with charge conjugation (the boundary is charge-neutral by [2]), the complement $(\mathbf{1} - \Pi)\Xi$ also satisfies the Majorana condition. The kernel spinor is therefore self-conjugate, globally neutral, and carries no independent charge sector. It is not a Dirac fermion and not a Weyl spinor. It is a Majorana object in the bulk that the boundary projection cannot transmit.

Physically: $\Xi_{\text{ker}}(163)$ is the part of the bulk that “tries to cross” but is reflected at the domain walls. It does not vanish: it organises itself into the internal structure $\mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$ and leaves evanescent traces on both sides of the wall.

The internal operator Γ_{163}

To make the internal structure precise, we introduce an intertwiner $J : \mathcal{B}_{13}^+ \rightarrow \mathcal{B}_{13}^-$ that represents the Majorana conjugation between the two chiral branches:

$$J = C|_{\mathcal{B}_{13}^+ \rightarrow \mathcal{B}_{13}^-}, \quad J^{-1}J = JJ^{-1} = \mathbf{1}_{13}. \quad (94)$$

From J we construct the internal operator:

$$\Gamma_{163} := \sqrt{163} \begin{pmatrix} 0 & J \\ -J^{-1} & 0 \end{pmatrix} \quad \text{on} \quad K_{163} := \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+. \quad (95)$$

Proposition A.2 (Minimal polynomial of Γ_{163}). $\Gamma_{163}^2 = -163 \cdot \mathbf{1}_{26}$. The normalised involution $A_{163} := i\Gamma_{163}/\sqrt{163}$ satisfies $A_{163}^2 = \mathbf{1}_{26}$, and the associated projectors $\Pi_{163}^\pm = \frac{1}{2}(\mathbf{1} \pm A_{163})$ are well-defined orthogonal projectors on K_{163} .

Proof. Direct computation: $\Gamma_{163}^2 = 163 \cdot \begin{pmatrix} 0 & J \\ -J^{-1} & 0 \end{pmatrix}^2 = 163 \cdot \begin{pmatrix} -JJ^{-1} & 0 \\ 0 & -J^{-1}J \end{pmatrix} = -163 \cdot \mathbf{1}_{26}$. The involution follows: $A_{163}^2 = (i/\sqrt{163})^2 \Gamma_{163}^2 = (-1/163) \cdot (-163 \cdot \mathbf{1}) = \mathbf{1}$. Idempotency of $\Pi_{163}^\pm$ is standard. $\square$

The number 163 enters not as an external input but as the *minimal polynomial* of an operator whose construction is forced by the Majorana structure of the bulk and the bipartition $\mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$. It is the first time a Heegner number appears as a derived algebraic invariant of a physical projection operator.

Pause here. 163 was a number observed in number theory — the largest Heegner discriminant, known since Gauss, proved by Baker and Stark. It is now realised in physics as the minimal-polynomial index of the Majorana transport operator, once the Baker–Stark terminus and the $\mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$ split are fixed: proved, but realised by construction rather than ab initio.

The minimal lock-in background

The kernel spinor lives against a background fixed by the lock-in. The background value is the QGT golden return — the first singular value σ_1 of the rank-5 projection and the Fibonacci pole, the fixed point of $R = 1/(1 + R)$:

$$R_\star := \mathcal{R}_{\text{QGT}} = \frac{1}{\varphi} = \varphi - 1, \quad (96)$$

from which the algebraic identity follows immediately:

$$\varphi^2 = 2 + \mathcal{R}_{\text{QGT}}. \quad (97)$$

This is a closed algebraic fact, not a conjecture: it follows from $R_\star = 1/\varphi$ and the golden equation $R_\star^2 + R_\star - 1 = 0$. (It is *not* the Rogers–Ramanujan value at $q = e^{-2\pi}$, which by Berndt’s evaluation [6, 7] is $R_{\text{RR}}(e^{-2\pi}) = \sqrt{(5 + \sqrt{5})/2} - \varphi \approx 0.284$; the two coincide only as the $q \rightarrow 1$ limit of the continued fraction.) The background amplitude of the lock-in soliton is therefore $A_\star = \varphi^2 = 2 + R_\star$, the unique value at which the modular seed and the boundary norm coincide.

The minimal lock-in solution compatible with the QGT boundary conditions of [3] (evanescent normal profile, $\partial_n \Psi_\star|_{\text{edge}} = 0$, $\xi_\star \eta_\star = \alpha^{-1}$, $P_\sigma \Psi_\star = 0$) is:

$$\Psi_\star(s, n) = \varphi \operatorname{sech}\left(\frac{n}{\ell_{\text{pen}}}\right) e^{i\omega_{\text{bal}} s} u_M, \quad u_M \in K_{163}, \quad (98)$$

where s is the tangential coordinate on $\Sigma : \xi\eta = \alpha^{-1}$, n is the normal coordinate, and $\omega_{\text{bal}} = 0.5284$ is the canonical balance frequency of [4]. The sech profile satisfies $\Psi_\star(0) \neq 0$, $\partial_n \Psi_\star(0) = 0$, and $\Psi_\star \rightarrow 0$ as $|n| \rightarrow \infty$: it is the minimal profile compatible with an evanescent edge mode that does not vanish at the wall. The amplitude $\varphi = \sqrt{A_\star}$ is fixed by the golden-return seed $R_\star = 1/\varphi$: no free parameter enters.

The linearised operator $L_\star^{(13)}$

Writing $\Psi = \Psi_\star + \varepsilon\chi$ and expanding the nonlinear boundary Dirac equation

$$(\gamma^\mu D_\mu + \alpha\xi + i\alpha\eta\gamma^5)\Psi + \lambda_{\text{eff}}|\Psi|^{2/3}\Psi = 0, \quad (99)$$

with $\lambda_{\text{eff}} = 16\pi^2\varphi^{12}$ and $\xi\eta = \alpha^{-1}$, one obtains the linearised operator $L_\star$. The block-diagonal structure

$$P_\sigma L_\star P_{\text{dyn}} = 0, \quad P_{\text{dyn}} L_\star P_\sigma = 0, \quad (100)$$

follows from the Mellin-diagonal action of $L_\star$ on the mode space: the constant mode P_σ (the pole sector $\sigma = 137$) is orthogonal to the dynamic sector $P_{\text{dyn}} = \mathbf{1} - P_\sigma$ under the Mellin inner product [3], so cross-coupling vanishes identically. This guarantees that $\sigma = 137$ is protected under linearisation.

On the internal branch $K_{163} = \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$, the linearised operator decomposes as:

$$L_\star^{(13)} = L_{\text{scale}}^{(5)} \oplus L_{\text{tens}}^{(8)}, \quad (101)$$

where $L_{\text{scale}}^{(5)} = \text{diag}(\lambda_0, \dots, \lambda_4)$ acts on S_5 through the five golden singular values, and

$$L_{\text{tens}}^{(8)} = J_{\xi\eta} \otimes I_4 + M_{\text{EW}}, \quad (102)$$

with $J_{\xi\eta}$ given by (36) and M_{EW} an 8×8 matrix that mixes the three boson channels (B_1, B_2, B_3) while leaving $(\alpha_\parallel, \alpha_\perp)$ normative.

The block form on $\mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$ is:

$$L_\star^{(163)} = \mathbb{H}_{13} + \frac{\varepsilon_{163}}{\sqrt{163}} \Gamma_{163}, \quad (103)$$

where $\mathbb{H}_{13} = \text{diag}(H_{13}^-, H_{13}^+)$ is the diagonal part and ε_{163} is the modular residue defined in the next section. The commutation condition $[L_\star^{(163)}, \Gamma_{163}] = 0$ requires $H_{13}^- J = J H_{13}^+$, i.e. the two 13-dimensional blocks are conjugate under J . This is satisfied by the explicit seed $\Psi_\star$ of Theorem 7.3 (Appendix C, Step 2): the intertwiner $J_\star u_\star = -iu_\star$ on the resonant subspace is the operator realisation of this conjugation.

The internal splitting

If λ_j is an eigenvalue of H_{13}^- , the eigenvalues of the full 2×2 block at level j are:

$$\lambda_j^{(\pm)} = \lambda_j \pm i\varepsilon_{163}. \quad (104)$$

For the balance mode $\lambda_j = \omega_{\text{bal}} = 0.5284$:

$$\omega_{\text{bal}}^{(\pm)} = 0.5284 \pm 0.0028. \quad (105)$$

Since $\varepsilon_{163} \approx 2.8 \times 10^{-3} \ll \omega_{\text{bal}} = 0.5284$, the kernel sector does not replace the boundary dynamics: it splits it internally. The domain walls sustain a doublet structure around every boundary mode, with the doublet gap set by the modular residue of the Ramanujan seed.

B The Ramanujan–Heegner layer

The kernel has a structure. But structure alone is not enough to say where it lives in the landscape of mathematics. This section gives the kernel its address: the arithmetic of imaginary quadratic fields. Four of the five admissible modes of the QGT are Heegner numbers. The fifth — the lock-in $n = 5$ — is the first prime to break the class-number-1 condition, and it is precisely there that the boundary acquires its observable form. The arithmetic is not decorative. It is the natural algebraic language in which the kernel architecture is encoded.

Established facts

The following are numerical and algebraic facts, not conjectures.

Proposition B.1 (Heegner–Admissible characterisation). *The admissible mode set $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ is characterised by the following Heegner–Baker–Stark criterion:*

$$\mathcal{N}_{\text{adm}} = \{p \text{ prime} : p \leq 11, h(-4p) = 1 \text{ or } p = 5\}, \quad (106)$$

where $h(-4p)$ is the class number of $\mathbb{Q}(\sqrt{-p})$, and the set $\{p : h(-4p) = 1\} = \{2, 3, 7, 11\}$ is determined by the Baker–Stark theorem (finite enumeration, no free parameter). The mode $n = 5$ is the unique prime excluded from $h(-4p) = 1$ but included in $\mathcal{N}_{\text{adm}}$: it is the first prime with $h(-20) = 2$, and it is precisely the lock-in mode where the boundary acquires its observable form. This characterisation is proved: all steps reduce to the Baker–Stark classification plus the QGT rank-5 selection $F_r = r \Rightarrow r = 5$.

(F1) The admissible modes satisfy $\mathcal{N}_{\text{adm}} \cap \mathcal{H} = \{2, 3, 7, 11\}$, where $\mathcal{H} = \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$ is the set of Heegner numbers (Baker–Stark theorem). Four of the five admissible modes are Heegner numbers. The fifth, $n = 5$, satisfies $h(-20) = 2$: it is the first prime to break the class-number-1 condition, and it is the lock-in.

(F2) The QGT golden return is $\mathcal{R}_{\text{QGT}} = 1/\varphi$, the Fibonacci pole and first singular value σ_1 of the rank-5 projection. (The Rogers–Ramanujan continued fraction reaches this value only as its $q \rightarrow 1$ limit; at $q = e^{-2\pi}$ its value is instead Berndt’s $R_{\text{RR}}(e^{-2\pi}) \approx 0.284$.) The golden return is the modular signature at the modulus of the QGT lock-in ($n = 5$, $\kappa^2 = 5$). Setting $R := \mathcal{R}_{\text{QGT}} = 1/\varphi = \varphi - 1$, the golden equation $R^2 + R - 1 = 0$ gives $R^2 = 1 - R$, and therefore:

$$\varphi^2 = (1 + R)^2 = 1 + 2R + R^2 = 1 + 2R + (1 - R) = 2 + R. \quad (107)$$

This is an exact algebraic identity, not an approximation:

$$\varphi^2 = 2 + \mathcal{R}_{\text{QGT}}. \quad (108)$$

The kernel spinor carries $R = 1/\varphi$ as its modular signature; the bulk Majorana spinor carries $\sqrt{5}$ as its Clifford spectral curvature [4]. These are connected by the unique algebraic identity that closes $\mathbb{Q}(\sqrt{5})$:

$$\varphi + \frac{1}{\varphi} = \sqrt{5}, \quad (109)$$

i.e. $\varphi + \mathcal{R}_{\text{QGT}} = \sqrt{5}$. The projection Π generates both simultaneously: the bulk spectral curvature ($\sqrt{5}$) and the kernel modular signature ($R = 1/\varphi$) are the two sides of the same closure identity. Neither can exist without the other.

(F3) The j -invariant at the CM point $\tau_{163} = (1 + i\sqrt{163})/2$ is $j(\tau_{163}) = -640320^3$ (complex multiplication theory). The near-integer identity $e^{\pi\sqrt{163}} = 640320^3 + 744 - \varepsilon$, $\varepsilon \approx 7.5 \times 10^{-13}$, is a consequence.

(F4) Using the golden identity $\varphi^2 = 2 + \mathcal{R}_{\text{QGT}}$,

$$e^{\pi\sqrt{163}} \approx \varphi^2 \cdot 10^{17} = (2 + \mathcal{R}_{\text{QGT}}) \cdot 10^{17}. \quad (110)$$

We stress that this is a base-10 numerical coincidence, not a derivation: the right-hand side is simply $\varphi^2 \cdot 10^{17}$, the factor 10^{17} carries no modular content, and $\mathcal{R}_{\text{QGT}}$ is the golden return, *not* the Rogers–Ramanujan value at $q = e^{-2\pi}$. The approximation $e^{\pi\sqrt{163}} \approx \varphi^2 \cdot 10^{17}$ is recorded as Corollary B.3 below and claims no arithmetic significance.

Two residuals measure the gap precisely. The *logarithmic residual*:

$$\varepsilon_{163} := \pi\sqrt{163} - \ln(\varphi^2 \cdot 10^{17}) = \ln\left(\frac{e^{\pi\sqrt{163}}}{\varphi^2 \cdot 10^{17}}\right) \approx 2.7998 \times 10^{-3}, \quad (111)$$

and the *relative residual*:

$$\delta_{163} := \frac{|e^{\pi\sqrt{163}} - \varphi^2 \cdot 10^{17}|}{e^{\pi\sqrt{163}}} \approx 2.7958 \times 10^{-3}. \quad (112)$$

These are distinct objects ($\varepsilon_{163} = \ln(1 + \delta_{163}/(1 - \delta_{163})) \approx \delta_{163}$ to leading order) but both satisfy the same window:

$$\frac{e}{10^3} < \delta_{163}, \varepsilon_{163} < \frac{2\sqrt{2}}{10^3}. \quad (113)$$

Numerically: $e/10^3 \approx 2.7183 \times 10^{-3}$, $\delta_{163} \approx 2.7958 \times 10^{-3}$, $\varepsilon_{163} \approx 2.7998 \times 10^{-3}$, $2\sqrt{2}/10^3 \approx 2.8284 \times 10^{-3}$. The fact that the same window captures both residuals — the multiplicative error and the additive logarithmic error — independently of the representation, is the structural content of the window: it is not an artefact of a particular normalisation, but a genuine feature of the gap between $e^{\pi\sqrt{163}}$ and $\varphi^2 \cdot 10^{17}$.

The lower bound $e/10^3$ is the entropic-logarithmic scale of the projection ([4] introduces e as the entropic projection scale, $\theta = \pi/e$, at which classical dissipation begins). The upper bound $2\sqrt{2}/10^3$ is the spinorial bilinear scale: $2\sqrt{2} = \sqrt{B^*}$ where $B^* = 8$ is the breathing attractor of the breathing attractor [4], and $2\sqrt{2}$ also appears as the prefactor in Ramanujan's series for $1/\pi$. The residual δ_{163} lives strictly between the entropic and spinorial registers of the boundary: it is too large to be purely entropic and too small to be purely spinorial. This is the arithmetic fingerprint of the kernel's position between the two boundary registers.

Corollary B.2 (The kernel window and the CMB temperature). *The observed CMB temperature satisfies:*

$$\frac{e}{10^3} < \frac{T_{\text{CMB}}}{10^3 \text{ K}} < \frac{2\sqrt{2}}{10^3}, \quad (114)$$

i.e. $T_{\text{CMB}}/10^3 \approx 2.7255 \times 10^{-3}$ falls inside the same entropic-spinorial window as δ_{163} and ε_{163} .

This common window is a Level-B structural observation. That the modular residuals δ_{163} , ε_{163} and the thermal ratio $T_{\text{CMB}}/10^3$ share the entropic-spinorial band $(e/10^3, 2\sqrt{2}/10^3)$ is recorded here as a structural readout; the thermal derivation and the harmonic realisation on the CMB sky are downstream of this kernel result and are not treated here.

A candidate thermal floor $T_0 = \pi/\ln \pi$ for the same $1/f$ structure is downstream of this kernel result (not derived here), with the lower spectral cutoff $f_{\min} = 1$ set by the Majorana 4π cycle and the Cayley fixed point $\theta = \pi$. That derivation belongs to the paper on the emergence of physical time; the kernel paper records the arithmetic signature of the same floor.

(F5) The cubic roots of the j -invariant at all five large Heegner discriminants satisfy: $c_d = |j((1 + i\sqrt{d})/2)|^{1/3} \equiv 0 \pmod{8}$ for $d \in \{11, 19, 43, 67, 163\}$. The breathing attractor $B^* = 8$ is therefore divisible into every c_d .

The Heegner index of the lock-in

Corollary B.3 (The arithmetic index of the kernel). *The arithmetic index of the kernel spinor $\Xi_{\ker}(163)$ is the largest Heegner number 163. It decomposes as:*

$$163 = 137 + 2 \times 13 = \sigma + \dim \mathcal{B}_{13}^{\text{cand}}, \quad (115)$$

where $\sigma = 137$ is the projection residue (proved in [3]), $26 = \dim K_{163} = \dim(\mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+)$ is the internal dynamic dimension (this paper, Theorems 7.3–7.5), and $13 = \dim \mathcal{B}_{13} = 5 + 8$ is the dimension of each chiral branch (Section 3).

The same 13 appears:

- (i) as $\dim \mathcal{B}_{13}$ (internal branch, this paper);
- (ii) as the denominator of $\sin^2 \theta_W = 3/13$ (proved in [3]);

(iii) as $2r + |\mathcal{C}_{\text{obs}}| = 2 \times 5 + 3$ (EW closure, [3]).

These are not three independent occurrences. They are three readings of the same number: the total capacity of the rank-5 tensor field with three dynamic boson channels, read from the boundary (mixing angle), from the lattice (closure count), and from the kernel (internal dimension).

The result is not numerology. The number 163 is the largest discriminant for which the ring of integers of $\mathbb{Q}(\sqrt{-163})$ is a principal ideal domain (class number 1, Baker–Stark [11, 12]). In $\mathcal{O}_{163} = \mathbb{Z}[(1 + \sqrt{-163})/2]$, the projection residue $\sigma = 137$ is a Gaussian prime [3]. The decomposition $163 = 137 + 26$ is not a sum of arbitrary integers: it is the partition of the maximum class-number-1 complexity into its observable residue (σ , a Gaussian prime in $\mathcal{O}_{163}$) and its internal dynamic dimension ($26 = 2 \times 13$, the bipartite kernel branch). With σ established in Appendix D and $\dim \mathcal{B}_{13} = 13$ fixed by the filter structure (Section 3), the discriminant 163 is the unique Heegner number that can simultaneously contain both.

The modular operator $\mathcal{M}_{\text{ker}}$

The full picture integrates into a single operator:

$$\mathcal{M}_{\text{ker}} : \ker \Pi \rightarrow \ker \Pi, \quad \text{Index}(\mathcal{M}_{\text{ker}}) = 163. \quad (116)$$

Its domain consists of the CM points $\tau_d = (1 + i\sqrt{d})/2$ for Heegner discriminants d : the operator “lives” in the arithmetic of imaginary quadratic fields. Its image on the boundary is the modular data that the boundary can read as shadows: $\mathcal{R}_{\text{QGT}} = 1/\varphi$ and $j(\tau_{163}) = -640320^3$.

In the QGT interpretation: $\mathcal{M}_{\text{ker}}$ is *Heegner in its domain and Ramanujan in its image*. The Heegner structure determines where the kernel is organised (the CM points in the upper half-plane). The Ramanujan structure determines what the boundary perceives of it (the modular values $R(q)$, the series for $1/\pi$, the near-integer $e^{\pi\sqrt{163}}$).

The two spinors of the QGT corpus are now both present and structurally located. The bulk Majorana spinor $\Xi = C\bar{\Xi}$ carries spectral curvature $\sqrt{5}$ under the Clifford dilatation of [4]. The kernel spinor $\Xi_{\text{ker}}(163)$ carries modular signature $\mathcal{R}_{\text{QGT}} = 1/\varphi$ at the lock-in point $\tau = i$. These two numbers are not independent: the identity $\varphi + 1/\varphi = \sqrt{5}$, equivalently

$$\varphi + \mathcal{R}_{\text{QGT}} = \sqrt{5}, \quad (117)$$

is the algebraic closure of $\mathbb{Q}(\sqrt{5})$. The projection Π cannot generate $\sqrt{5}$ in the bulk without simultaneously placing $R = 1/\varphi$ in the kernel modular signature, because $\sqrt{5}$ and $1/\varphi$ are the two expressions of φ under the field automorphism $\varphi \mapsto -1/\varphi$ of $\mathbb{Q}(\sqrt{5})$. They are not two observations that happen to fit: they are one algebraic object read from two sides of the projection.

Remark B.1 (The two spinors and φ). *The bulk spinor and the kernel spinor are two projections of the same φ :*

$$\underbrace{\sqrt{5}}_{\text{bulk Clifford curvature}} = \varphi + \underbrace{\mathcal{R}_{\text{QGT}}}_{\text{kernel modular signature}}. \quad (118)$$

A rank-5 Majorana projection with Clifford curvature $\sqrt{5}$ must produce a kernel whose modular signature at the lock-in reads $R = 1/\varphi$, because $\sqrt{5} - \varphi = 1/\varphi$ is an identity in $\mathbb{Q}(\sqrt{5})$, not a coincidence. The dynamical content of this pairing — the Mellin-diagonal action of $L_\star$ on $\mathcal{B}_{13}$ — is established by Theorem 7.3 and its analytic counterpart Theorem 7.4 in Appendix C. The identity $\varphi + \mathcal{R}_{\text{QGT}} = \sqrt{5}$ is the arithmetic seal of their equivalence.

The connection to the Heegner index is the content of (F4) above: $e^{\pi\sqrt{163}} \approx (2 + R) \cdot 10^{17}$ places the maximum arithmetic depth 163 in the same formula as the kernel’s modular signature $R = 1/\varphi$. The residual $\delta_{163} \in (e/10^3, 2\sqrt{2}/10^3)$ measures how far τ_{163} is from exact coincidence

of the two registers — a coincidence that would require $\pi \in \mathbb{Q}(\sqrt{5})$, forbidden by Lindemann–Weierstrass. The gap is not a defect: it is what separates the two spinors structurally, and the temporal ordering established in [4] is its physical reading.

In the QGT interpretation, Ramanujan’s CM identities occupy the same analytic register that the theory identifies with the anti-causal/kernel component. The modular forms, the Rogers–Ramanujan fractions, the near-integers $e^{\pi\sqrt{d}}$ live at the $n = 0$ mode — total decoherence, all admissible filters inactive — where the kernel $\ker \Pi$ is directly readable and its grammar is modular. Read this way, they are values of H_- , the anti-causal component of the Pincherle filter, at CM points in the upper half-plane: the imaginary part of a holomorphic function is fixed by its real part via the Hilbert transform, and the values at CM points by complex multiplication. This is an interpretive reading, not a literal historical or physical claim. Hardy, at the boundary, received the real-valued formulas and verified them by L^2 analysis, without seeing their origin. The Hardy–Ramanujan collaboration is, in this reading, the Hilbert decomposition enacted in historical time: the causal part (boundary, real, verifiable by analysis) and the anti-causal part (kernel, imaginary, accessible by complex multiplication) were held by two mathematicians who never shared the same register.

C Proofs of Theorems 7.3–7.5 and Proposition 7.6

Section 7: the dynamical invariance of $\mathcal{B}_{13}$, the analytic closure at τ_{163} , the equivalence of the two formulations, and the Noether-norm derivation of λ_{lock} . The proofs use only the structures established in the main text.

A1. Dynamical invariance: $L_*(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$

Proof of Theorem 7.3. Step 1 — Explicit seed. Let

$$\Psi_*(n, s, \theta) = \varphi \operatorname{sech}(n/\ell_{\text{pen}}) e^{i(\omega_{\text{bal}}s + \Gamma_{\tau}\theta)} u_*,$$

with $u_* = (s_* + \tau_*)/\sqrt{2} \in \mathcal{B}_{13}$ the normalised combination of the scale generator $s_* \in S_5$ and the tensor generator $\tau_* \in \mathcal{H}_* \subset T_8$. The penetration length is ℓ_{pen} (setting the edge-mode decay scale, [3]), and ω_{bal} is the balance frequency of the lock-in. Ψ_* satisfies $\partial_n \Psi_*|_{\text{edge}} = 0$ (sech is even, with $\partial_n \operatorname{sech} = 0$ at $n = 0$) and $P_{\sigma} \Psi_* = 0$ (the spectral pole mode is not excited).

Step 2 — Majorana intertwiner on the core. The resonant intertwiner J_* acts on $\mathcal{H}_*$. By the internal operator Γ_{163} (Section 4) one has $J_*^2 = -\mathbf{1}$ and $[L_{\text{core}}, J_*] = 0$, so $J_* u_* = -i u_*$ on the one-dimensional resonant subspace generated by u_* .

Step 3 — Mellin diagonality under the lock-in ansatz. The power-law nonlinearity $\lambda|\Psi|^{2/3}\Psi$ induces a Mellin convolution in general, not a diagonal action (Proposition 4.3). Under the Mellin-diagonal lock-in ansatz — that the projected lock-in seed is Mellin-diagonal on the retained basis — the projected Fréchet linearisation L_* acts diagonally on the retained $\mathcal{B}_{13}$ basis, as a scalar multiplication on each Mellin mode.

Step 4 — Invariance of $\mathcal{B}_{13}$. The branch $\mathcal{B}_{13} = S_5 \oplus T_8$ is the span of the first $\dim \mathcal{B}_{13} = 13$ Mellin modes of the filtered kernel (Section 5): five from the polynomial P_4 , eight from the tensor structure at the pole. Any Mellin-diagonal operator preserves any span of Mellin modes; in particular, $L_*(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$.

Step 5 — Spectral dimension. The Majorana condition doubles the internal space into two chiral branches $H_{13}^{\pm}$, giving $\dim \mathcal{H}_{\text{dyn}} = \dim(H_{13}^+ \oplus H_{13}^-) = 26$. This is a spectral output, not an a priori count. $\square$ $\square$

A2. Analytic closure at τ_{163}

Proof of Theorem 7.4. Step 1 — Filter is scalar at τ_{163} . The nome at the maximum Heegner point is $|q_{163}| = e^{-\pi\sqrt{163}}$. With $H(z) = P_4(z)/(1 - \alpha z)$, the filter at $z = e^{2\pi i\tau_{163}}$ becomes

$$H(e^{2\pi i\tau_{163}}) = \sigma_0 \left(1 + \varepsilon_{163}^{(\text{mod})} \right),$$

with a correction $\varepsilon_{163}^{(\text{mod})}$ of order $\alpha e^{-\pi\sqrt{163}}$, which is arithmetically negligible at any structural precision of interest. At the stated precision, H acts as the pure scalar σ_0 .

Step 2 — Anti-causal branch. By Step 2 of the proof of Theorem 7.3, $J_\star u_\star = -iu_\star$ on the resonant subspace. The Majorana condition $\Xi_-^\star = C\Xi_+^\star \simeq -i\mathcal{H}[\Xi_+^\star]$ identifies the anti-causal projection $P^-\Xi_\star$ with a vector proportional to $u_\star$ in H_{13}^- :

$$P^-\Xi_\star = \Xi_-^\star \in H_{13}^- \subset \mathcal{B}_{13}.$$

Step 3 — Scalar commutes with P^- . Since $H|_{\tau_{163}} = \sigma_0 \cdot 1$,

$$P^-(H(e^{2\pi i\tau_{163}})\Xi_\star) = \sigma_0 P^-\Xi_\star = \sigma_0 \Xi_-^\star \in H_{13}^- \subset \mathcal{B}_{13}.$$

□

□

Remark C.1. *The content of the proof is that the transcendence of $e^{\pi\sqrt{163}}$ suppresses all modular structure of the filter at τ_{163} , leaving only the scalar leading term. The CM point τ_{163} is the unique point in $\mathbb{H}$ where the filter “forgets its shape and sees only its pole”. This is why the analytic path is shorter than the operator path: arithmetic suppresses what differentiation would have to diagonalise.*

A3. Equivalence of the two formulations

Proof of Theorem 7.5. Forward direction ($L_\star$ -invariance $\Rightarrow$ analytic closure). Given $L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$ and $\Psi_\star$ the explicit seed of Theorem 7.3, the Mellin diagonality (Step 3 of the proof of Theorem 7.3) implies that $\mathcal{H}[\Psi_\star] \in \mathcal{B}_{13}$ because the Hilbert transform is a Mellin multiplier by $-i \text{sgn}$ on each mode and $\mathcal{B}_{13}$ is Mellin-closed. Combined with the Majorana identification $\Xi_\star$ of Step 2 and the scalar action of H at τ_{163} (Step 1 of the proof of Theorem 7.4), one gets $P^-(H(e^{2\pi i\tau_{163}})\Xi_\star) \in \mathcal{B}_{13}$.

Backward direction (analytic closure $\Rightarrow L_\star$ -invariance). Given the analytic closure, the Hilbert transform of $\Psi_\star$ projects into $H_{13}^- \subset \mathcal{B}_{13}$. Since the Hilbert transform is Mellin-diagonal (acts as $-i \text{sgn}$ per mode) and the filtered kernel basis is exactly the first 13 Mellin modes (Section 5), any operator that commutes with Mellin dilatation — $L_\star$ in particular — maps this span to itself. Hence $L_\star(\mathcal{B}_{13}) \subseteq \mathcal{B}_{13}$. □

A4. Noether-norm derivation of λ_{lock}

Proof of Proposition 7.6. Step 1 — The invariant surface. By Noether conservation, $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$ on every physical orbit, so the invariant surface is $\Sigma = \{(\xi, \eta) : \xi\eta = \alpha^{-1}\}$.

Step 2 — Symmetry group and Haar measure. Parametrise Σ by $t = \log \xi$: $\xi = \sqrt{\alpha^{-1}} e^t$, $\eta = \sqrt{\alpha^{-1}} e^{-t}$, $t \in \mathbb{R}$. The symmetry group is $(\mathbb{R}, +)$; its Haar measure is the Lebesgue measure dt .

Step 3 — Ergodic mean. The mean of $\xi\eta$ over Σ is

$$\mathbb{E}_\Sigma[\xi\eta] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \xi(t)\eta(t) dt = \alpha^{-1},$$

since $\xi\eta = \alpha^{-1}$ is constant on Σ . Hence $\mathbb{E}_\Sigma[\xi\eta] = \alpha^{-1}$.

Step 4 — Coupling to the monodromy cycle. The resonant sector is coupled to the monodromy cycle of spectral norm $\mathcal{S}$ (the monodromy step count, [3]), which normalises the amplitude of the lock-in solution. The dual identity $\alpha^{-1} = z_{\text{pole}} = \mathcal{K}_\tau$ (projection pole equals Noether charge) ensures that the norm on $\mathcal{H}_*$ and the pole of $H(z)$ measure the same quantity. The resulting coupling constant is

$$\lambda_{\text{lock}} = \frac{\sqrt{\mathbb{E}_\Sigma[\xi\eta]}}{\mathcal{S}} = \frac{\sqrt{\alpha^{-1}}}{\mathcal{S}} = \sqrt{\frac{\alpha^{-1}}{\mathcal{S}^2}} = \sqrt{\Upsilon_{\text{geom}}}.$$

Using the winding form $\alpha^{-1} = n_{\min}^2 r \varphi^4$ [3] (with $n_{\min}$ the spinorial covering degree and r the projection rank), together with $\mathcal{S} = n_{\min} r$, one obtains $\Upsilon_{\text{geom}} = \varphi^4/r$ and

$$\lambda_{\text{lock}} = \frac{\varphi^2}{\sqrt{r}} = \frac{\varphi^2}{\sqrt{5}} = \Gamma_\tau \varphi.$$

□

□

Remark C.2 (Thermal-damping identity). *Proposition 7.6 has a direct consequence for the kernel trace on the sky. The thermal damping scale $\ell_T = \sigma T_{\text{CMB}}/(\Gamma_\tau \pi)$, evaluated using $T_{\text{CMB}} = (\pi/\ln \pi)(1 - 1/(\sqrt{2} \cdot 100))$, satisfies*

$$\ell_T = \alpha^{-1} + 2 \dim \mathcal{B}_{13} + \Upsilon_{\text{geom}} = 163 + \lambda_{\text{lock}}^2. \quad (119)$$

The arithmetic kernel index and the Noether coupling squared are the two invariants whose sum fixes the thermal damping. This identity closes the chain kernel index $\rightarrow$ lock-in coupling $\rightarrow$ CMB damping on a single equation. The harmonic structure it generates is downstream of this kernel result and is not treated here.

D Compressed recalls of imported structural results

The main text imports five results from companion papers. This appendix derives each one in compressed form so that the paper stands alone. The citations are retained for credit and for the reader who wants the full treatment.

The projection residue $\sigma = 137$ is the integer left over after the rank-5 projection Π absorbs as much of the winding norm $(5 + 3\sqrt{5})^2$ as possible in multiples of the elementary block $\beta = (n_{\min} \cdot n_3)^2 = 36$. The decomposition is $\alpha^{-1} = r\beta + \sigma$, with $r = 5$ and σ required to be prime, larger than β , and unique (rank-one hypothesis: no second prime of the same form may exist). Writing $r_k = (5 + 3\sqrt{5})^2 - k\beta$ for $k = 1, \dots, 8$, the only entry satisfying all three conditions is $k = 5$: $r_5 = 317 - 180 = 137$, prime and greater than 36. The candidates $k = 1$ and $k = 4$ produce primes 281 and 173 but both fail rank-one uniqueness, since their distances from 137 are multiples of β , which would require a second independent generator. Therefore $\sigma = 137$ is the unique solution.

The Noether invariant $\xi\eta = \alpha^{-1}$ follows directly from the single conservation law $\partial_I J^I = 0$ on $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$. Along the hyperbolic breathing $\dot{\xi} = -\Gamma_\tau \xi$, $\dot{\eta} = +\Gamma_\tau \eta$, the product $\mathcal{K}_\tau := \xi\eta$ is conserved. Its value at the lock-in $n_* = 5$ is fixed by the winding form $\alpha^{-1} = n_{\min}^2 \cdot r \cdot \varphi^4 = 20\varphi^4$ (five SVD modes, spinorial covering degree $n_{\min} = 2$); therefore $\xi\eta = \alpha^{-1}$ on every physical orbit. The product of the two chiral amplitudes equals the fine-structure constant inverse, conserved by the exchange symmetry of the temporal fibre.

The golden SVD hierarchy $\sigma_k \propto \varphi^{-k}$ follows from the spectral radius of the Pincherle dilatation $D_\tau f(x) = f(\varphi x)$ on the compact fibre. By the Fibonacci recursion $\varphi^2 = \varphi + 1$, φ is the Perron–Frobenius eigenvalue of the golden substitution matrix, so the retained modes decay as $\sigma_k \propto \varphi^{-k}$ for $k = 0, \dots, 4$. The five coefficients of $P_4(z) = \sum_{k=0}^4 \sigma_k z^k$ are therefore fixed by the Fibonacci structure with no free parameters.

The balance frequency $\omega_{\text{bal}} = 0.5284$ governs fluctuations $(\delta\xi, \delta\eta)$ tangent to the Noether surface $\Sigma : \xi\eta = \alpha^{-1}$. Setting $u := \delta\xi/\xi^* = -\delta\eta/\eta^*$, the linearised exchange equations give $\ddot{u} + \omega_{\text{bal}}^2 u = 0$ with $\omega_{\text{bal}}^2 = C_\kappa/C_\mu$, where $C_\kappa = \Gamma_\tau^2 \cdot \alpha^{-1}/(\xi^*\eta^*)$ and $C_\mu = \Gamma_\tau \cdot 146/55$ is the balance ratio of the $\mathbb{Z}_{10}$ monodromy cycle. Since $\xi^*\eta^* = \alpha^{-1}$, α cancels exactly and $\omega_{\text{bal}}^2 = \Gamma_\tau \cdot 146/55$ depends only on the golden connection $\Gamma_\tau = \varphi/\sqrt{5}$ and the rational $146/55$. The balance frequency is independent of the fine-structure constant; numerically $\omega_{\text{bal}} = 0.5284$.

The decoherent floor $n = 0$ (all admissible filters inactive) sets the entropic–spinorial window of the kernel; the measured CMB temperature falls within this window. This is recorded as a Level-B compatibility, without derivation: the thermal floor is downstream of the present kernel result and is not derived in this paper.

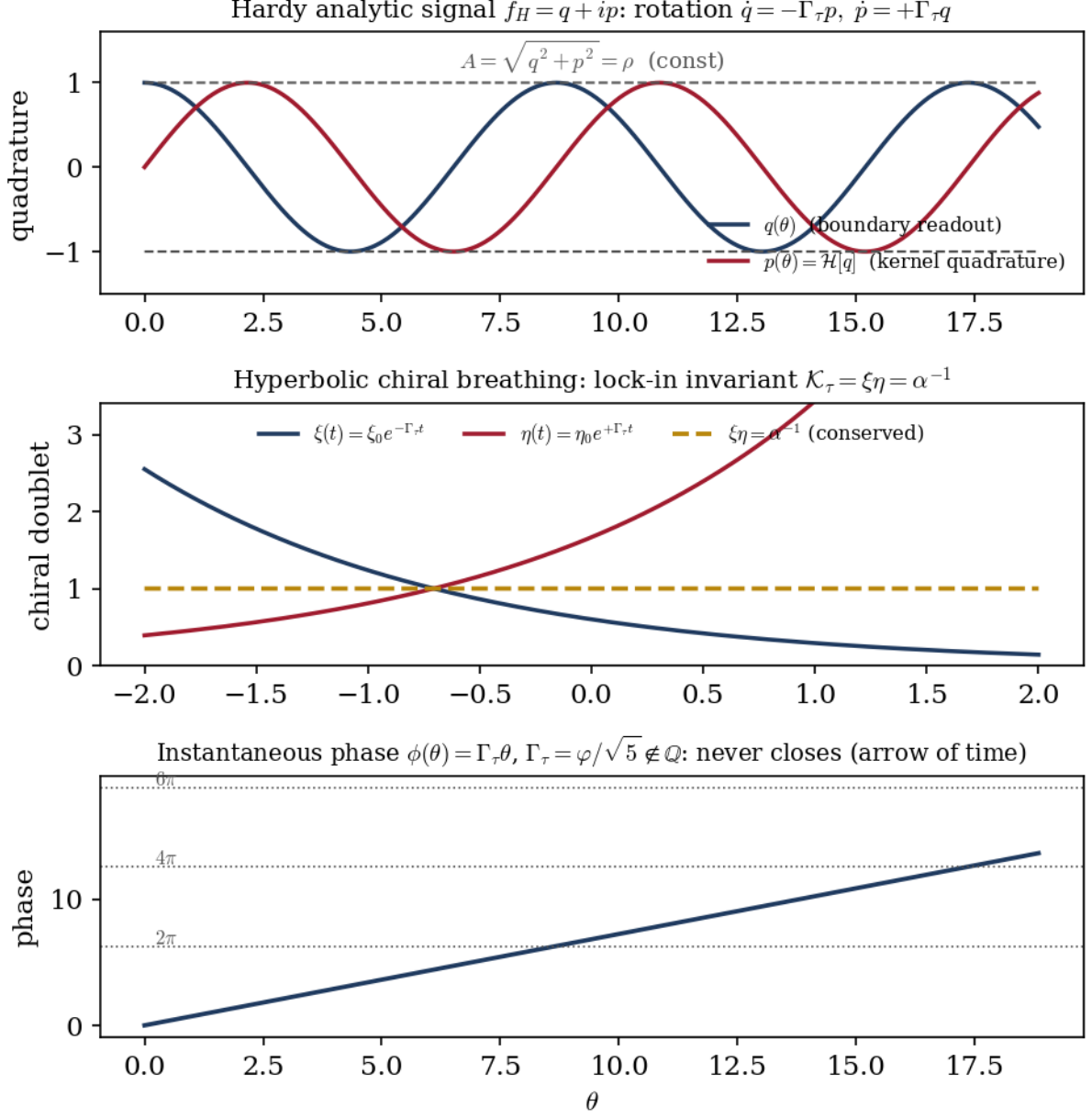


Figure 1: Analytic signal structure on $S^1(\tau)$. *Top:* the two quadrature components of the Hardy analytic signal, $q(\theta)$ (boundary readout, blue) and $p(\theta) = \mathcal{H}[q]$ (kernel quadrature, red), in permanent quadrature; the envelope $A(\theta) = \sqrt{q^2 + p^2}$ is their instantaneous amplitude. *Middle:* the lock-in invariant $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$ is conserved to numerical precision — the Noether invariant of the hyperbolic chiral breathing $\dot{\xi} = -\Gamma_\tau \xi$, $\dot{\eta} = +\Gamma_\tau \eta$. *Bottom:* the instantaneous phase $\phi(\theta) = \Gamma_\tau \theta$ accumulates at the golden rate $\Gamma_\tau = \varphi/\sqrt{5}$ but never closes ($\Gamma_\tau \notin \mathbb{Q}$): this is the arrow of time.

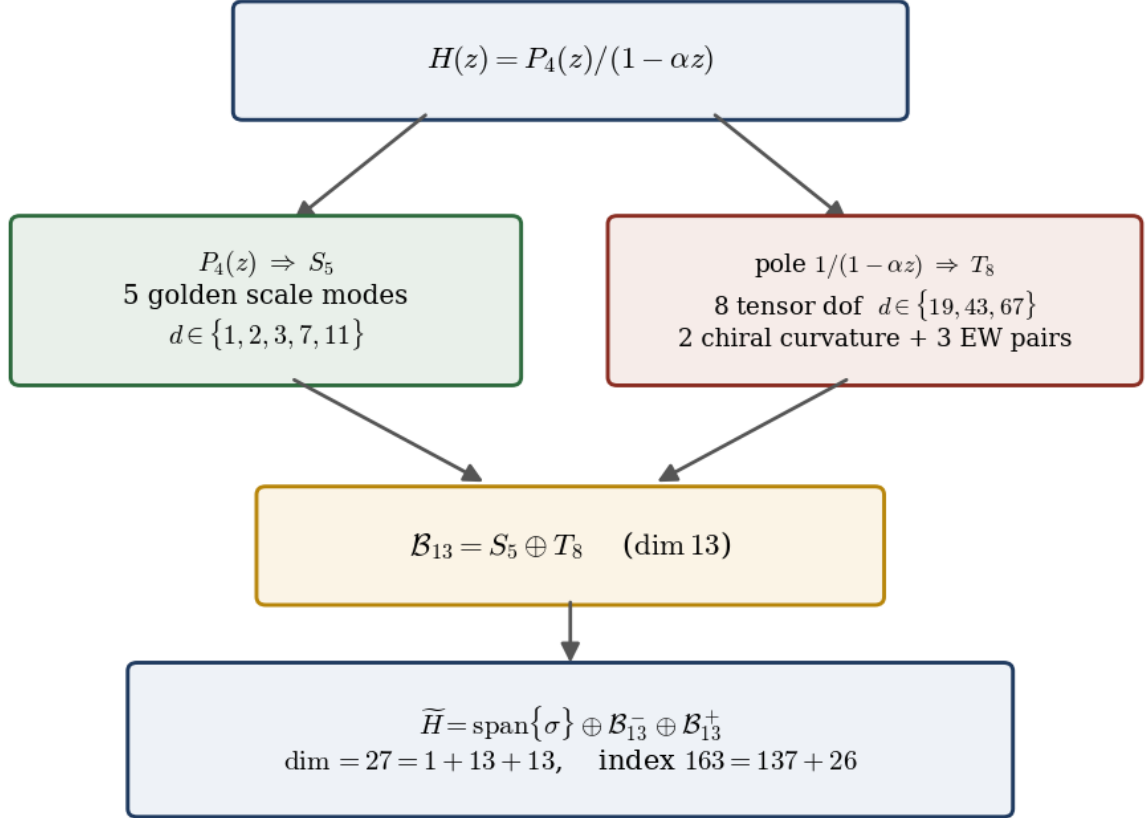
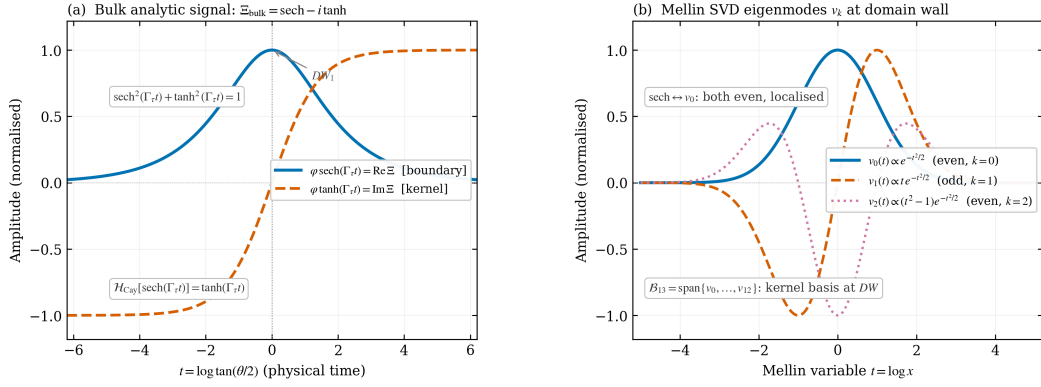


Figure 2: Internal structure $\mathcal{B}_{13} = S_5 \oplus T_8$ derived from the filter $H(z) = P_4(z)/(1 - \alpha z)$. The polynomial P_4 generates the five golden scale modes (S_5); the pole $1/(1 - \alpha z)$ at the anisotropic lock-in generates the eight tensor degrees of freedom (T_8), corresponding to the two chiral curvature channels and the three electroweak boson pairs. The total internal space $\tilde{H} = \text{span}\{\sigma\} \oplus \mathcal{B}_{13}^- \oplus \mathcal{B}_{13}^+$ has dimension $27 = 1 + 13 + 13$, with arithmetic index $163 = 137 + 26$.



Both Ψ_* and v_0 are even, localised, and decay monotonically. v_1 and v_2 carry odd/quadratic corrections. The structural equivalence $\text{sech} \leftrightarrow v_0$ justifies $\mathcal{B}_{13} = \text{span}\{v_0, \dots, v_{12}\}$ as the natural basis of $\ker \Pi$ at the domain walls.

Figure 3: *Left:* the lock-in profile $\Psi_*(n) = \varphi \text{sech}(n/\ell_{\text{pen}})$, normalised. It is even, localised, flat at $n = 0$ ($\partial_n \Psi_*|_0 = 0$), and decays exponentially into the bulk. *Right:* the first three Mellin SVD eigenmodes $v_k(t)$, $t = \log x$, with Gaussian envelope. v_0 (even, localised, monotone decay) and the sech profile have the same structural form: both are the unique even, unit-norm, rapidly decaying function in their respective domains. v_1 and v_2 carry the odd and quadratic corrections. This structural equivalence $\text{sech} \leftrightarrow v_0$ is the geometric justification for identifying $\mathcal{B}_{13} = S_5 \oplus T_8$ as the span of the first 13 Mellin modes of the filtered kernel: the lock-in soliton and the Mellin basis inhabit the same function space, read from the normal and the logarithmic directions respectively.

$H(z) = P_4(z)/(1 - \alpha z)$ on the Riemann sphere $\mathbb{P}^1$

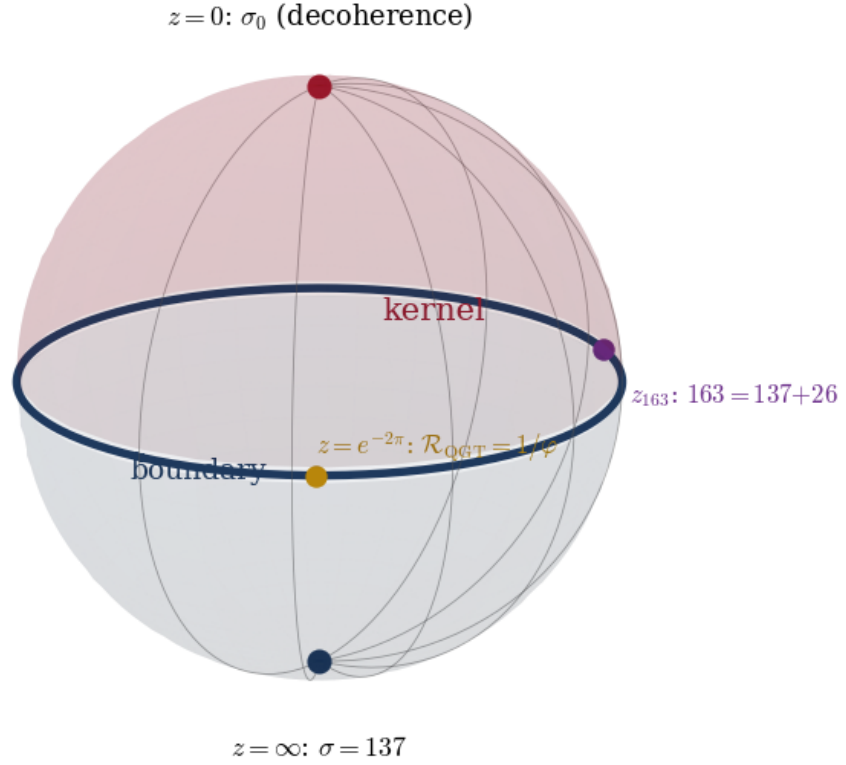


Figure 4: The filter $H(z) = P_4(z)/(1 - \alpha z)$ on the Riemann sphere $\mathbb{P}^1 = \mathbb{C} \cup \{\infty\}$. The unit circle (equator) is the observable boundary. The northern hemisphere ($|z| < 1$) is the kernel. The north pole $z = 0$ carries the zero mode σ_0 (complete decoherence); the south pole $z = \infty$ carries the projection residue $\sigma = 137$. The projection $\Pi = \text{Re}[\cdot]$ (in the analytic-signal chart) extracts the real part on the equator. The conformal map $z = e^{2\pi i \tau}$ continues the equator into the upper half-plane: the point $z = e^{-2\pi}$ ($\tau = i$) reads the golden frequency $\mathcal{R}_{\text{QGT}} = 1/\varphi$; the point $z_{163} = -e^{-\pi\sqrt{163}}$ (τ_{163}) reads the maximum Heegner structure with $163 = 137 + 26$. The two languages — Fibonacci (equator) and modular (northern hemisphere) — are the same function read from two sides.

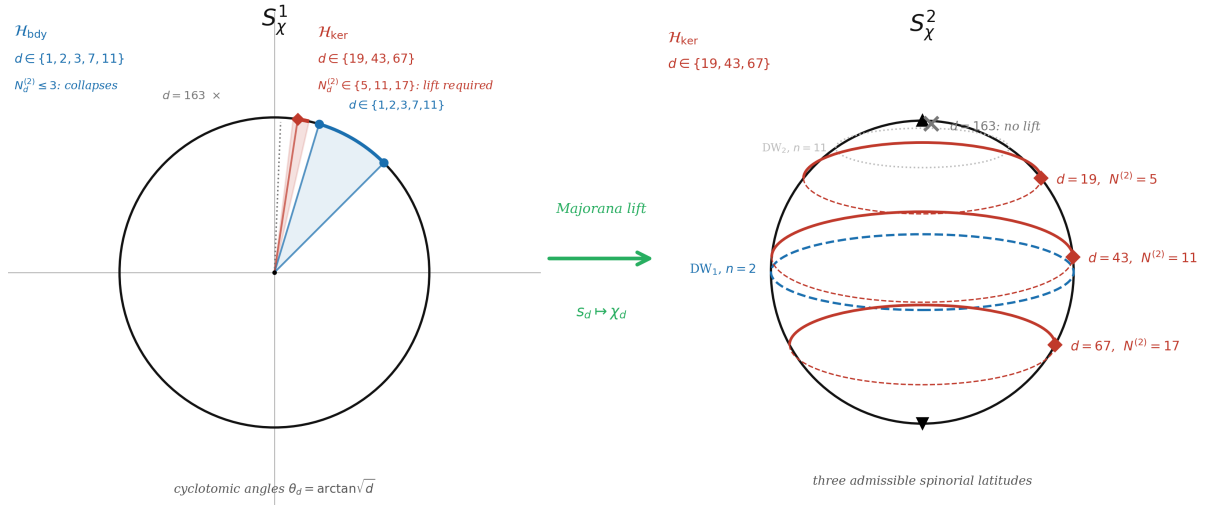


Figure 5: The Heegner phasor partition: cyclotomic collapse on S^1_χ and spinorial lift to S^2_χ . **Left** ($\mathcal{H}_{\text{body}}$, $d \in \{1, 2, 3, 7, 11\}$): each field defines a phasor $\rho_d = (1 + i\sqrt{d})/(2\sqrt{N_d^{(2)}}) \in U(1)$ at angle $\theta_d = \arctan \sqrt{d}$. The second norm $N_d^{(2)} \in \{2, 3\}$ is fixed on the two smallest primes regardless of d : the phasor perceives only angle, never depth, and its holonomy closes cyclotomically on S^1_χ . These are the boundary modes; Π carries them to $\mathbb{R}^4$. **Right** ($\mathcal{H}_{\text{ker}}$, $d \in \{19, 43, 67\}$): the second norm $N_d^{(2)} \in \{5, 11, 17\}$ grows as $d/4$. The phasor acquires spinorial depth and cannot close on S^1_χ ; the Majorana self-conjugation condition forces the lift $s_d \mapsto \chi_d = (s_d, \bar{s}_d)^\top \in S^2_\chi$. Each field opens one kernel generation at a distinct latitude on S^2_χ , with condensation eigenvalue $\lambda_d \propto N_d^{(2)}$. **Terminus** ($d = 163$): $N_{163}^{(2)} = 41 > n_{\text{sat}} = 11$; no admissible mode can sustain a spinorial step of size 41. The trajectory has no lift inside the projection Π . Baker and Stark proved that no further class-number-one field exists beyond $d = 163$ [11, 12]; the QGT reads this as the closure of the generation sequence. The partition $\mathcal{B}_{13} = S_5 \oplus T_8$ is the direct image of this geometric bifurcation.